

Definite Integration

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Integration (ref i2)

This document contains all 70 *Definite Integration* questions, each with a fully worked solution and a TikZ graph. A definite integral measures the signed area between the curve and the x -axis over the given limits, so each diagram shades that region: **green** where the curve is above the axis and **orange** where it is below (illustrating signed vs total area), with **blue** for the area trapped between two curves. The integration limits are shown as dashed lines.

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Definite Integration 01 (i2-001)

Difficulty: Foundation / Marks: 3

Evaluate $\int_1^3 (2x + 5) \, dx$.

Worked solution.

Step 1: Integrate term by term.

$$\int (2x + 5) \, dx = x^2 + 5x$$

Apply the power rule to each term. No $+C$ is needed for a definite integral — if we wrote $F(x) + C$ then the $+C$ would cancel in the subtraction $F(b) - F(a)$, so it never affects the answer.

Step 2: Apply the upper limit $x = 3$.

$$(3)^2 + 5(3) = 9 + 15 = 24$$

Substitute $x = 3$ into the integrated expression.

Step 3: Apply the lower limit $x = 1$.

$$(1)^2 + 5(1) = 1 + 5 = 6$$

Substitute $x = 1$ into the integrated expression.

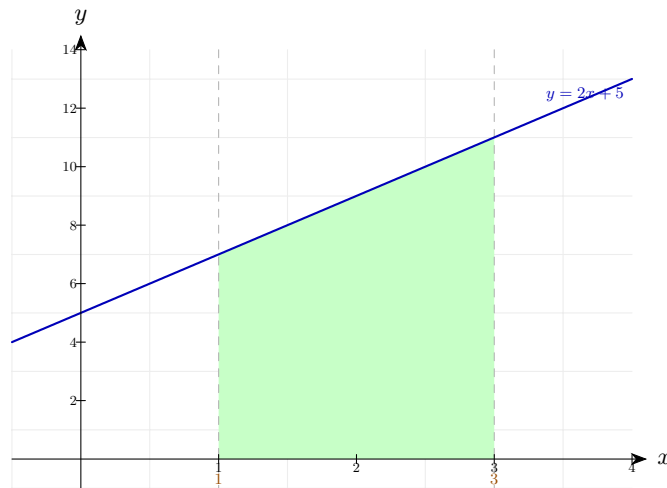
Step 4: Subtract lower from upper.

$$24 - 6 = 18$$

The definite integral equals $F(b) - F(a)$, upper minus lower. Reversing the order would flip the sign, so always subtract the lower-limit value from the upper-limit value.

Final answer: 18

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 02 (i2-002)

Difficulty: Foundation / Marks: 3

Evaluate $\int_0^2 (3x^2 - 4x + 1) \, dx$.

Worked solution.

Step 1: Integrate term by term.

$$\int (3x^2 - 4x + 1) \, dx = x^3 - 2x^2 + x$$

Apply the power rule to each term.

Step 2: Apply $x = 2$.

$$(2)^3 - 2(2)^2 + 2 = 8 - 8 + 2 = 2$$

Upper limit substitution.

Step 3: Apply $x = 0$.

$$0 - 0 + 0 = 0$$

Lower limit substitution gives zero.

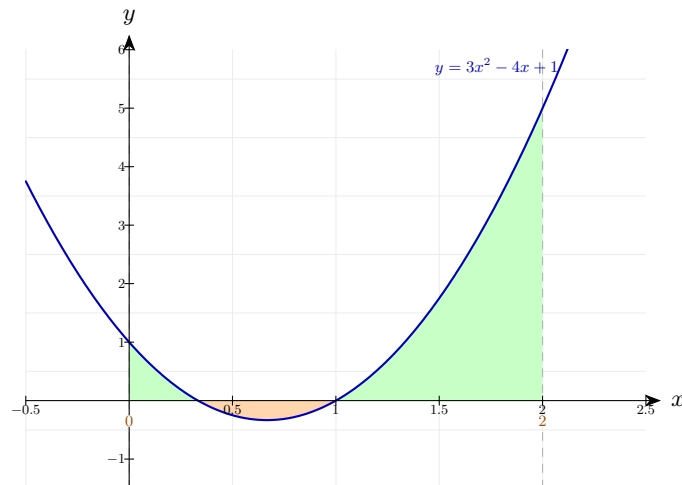
Step 4: Subtract.

$$2 - 0 = 2$$

The value of the definite integral.

Final answer: 2

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 03 (i2-003)

Difficulty: Foundation / Marks: 3

Evaluate $\int_1^4 \sqrt{x} \, dx$, giving an exact answer.

Worked solution.

Step 1: Write $\sqrt{x} = x^{1/2}$ and integrate.

$$\int x^{1/2} \, dx = \frac{x^{3/2}}{3/2} = \frac{2}{3}x^{3/2}$$

Use the power rule with $n = \frac{1}{2}$.

Step 2: Apply $x = 4$.

$$\frac{2}{3}(4)^{3/2} = \frac{2}{3} \times 8 = \frac{16}{3}$$

$$4^{3/2} = (\sqrt{4})^3 = 2^3 = 8.$$

Step 3: Apply $x = 1$.

$$\frac{2}{3}(1)^{3/2} = \frac{2}{3}$$

$$1^{3/2} = 1.$$

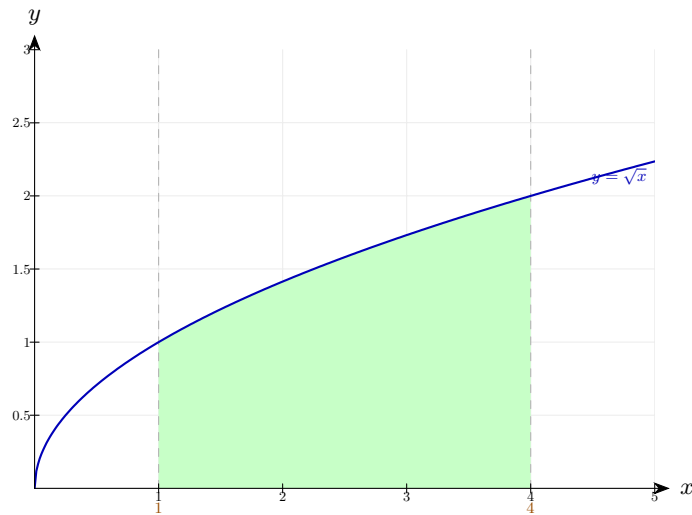
Step 4: Subtract.

$$\frac{16}{3} - \frac{2}{3} = \frac{14}{3}$$

Exact value of the integral.

Final answer: $\frac{14}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 04 (i2-004)

Difficulty: Foundation / Marks: 3

Evaluate $\int_1^2 \frac{4}{x^2} dx$.

Worked solution.

Step 1: Rewrite $\frac{4}{x^2} = 4x^{-2}$ and integrate.

$$\int 4x^{-2} dx = \frac{4x^{-1}}{-1} = -\frac{4}{x}$$

Apply the power rule with $n = -2$.

Step 2: Apply $x = 2$.

$$-\frac{4}{2} = -2$$

Upper limit substitution.

Step 3: Apply $x = 1$.

$$-\frac{4}{1} = -4$$

Lower limit substitution.

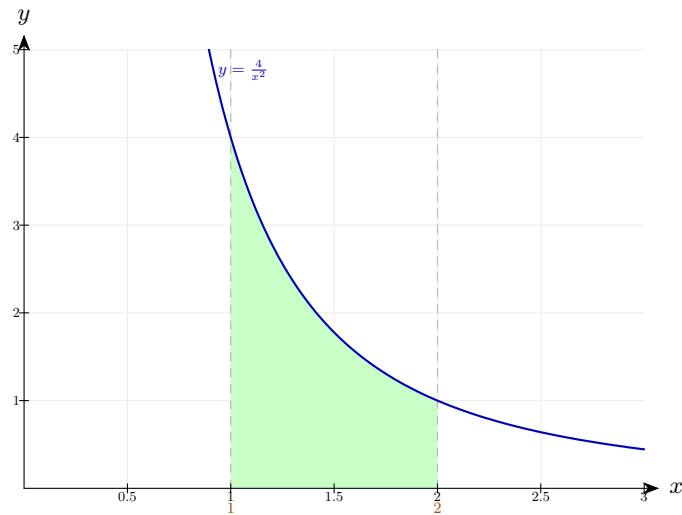
Step 4: Subtract.

$$-2 - (-4) = 2$$

Upper minus lower limit value.

Final answer: 2

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 05 (i2-005)

Difficulty: Foundation / Marks: 3

Evaluate $\int_{-1}^2 (x^2 + 3) \, dx$.

Worked solution.

Step 1: Integrate.

$$\int (x^2 + 3) \, dx = \frac{x^3}{3} + 3x$$

Power rule applied to each term.

Step 2: Apply $x = 2$.

$$\frac{8}{3} + 6 = \frac{8}{3} + \frac{18}{3} = \frac{26}{3}$$

Upper limit gives $\frac{26}{3}$.

Step 3: Apply $x = -1$.

$$\frac{(-1)^3}{3} + 3(-1) = -\frac{1}{3} - 3 = -\frac{10}{3}$$

Take care with negatives when cubing.

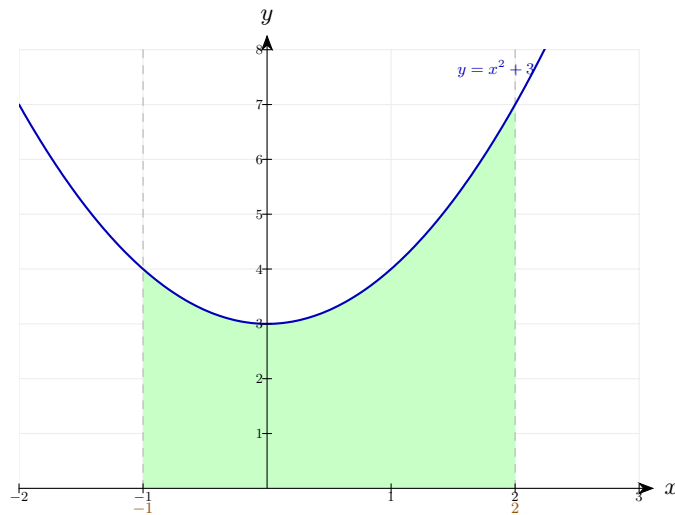
Step 4: Subtract.

$$\frac{26}{3} - \left(-\frac{10}{3}\right) = \frac{36}{3} = 12$$

Subtracting a negative adds the value.

Final answer: 12

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 06 (i2-006)

Difficulty: Foundation / Marks: 3

Evaluate $\int_0^3 (4x^3 - 6x) \, dx$.

Worked solution.

Step 1: Integrate.

$$\int (4x^3 - 6x) \, dx = x^4 - 3x^2$$

Apply the power rule to each term.

Step 2: Apply $x = 3$.

$$(3)^4 - 3(3)^2 = 81 - 27 = 54$$

Upper limit substitution.

Step 3: Apply $x = 0$.

$$0 - 0 = 0$$

Lower limit substitution.

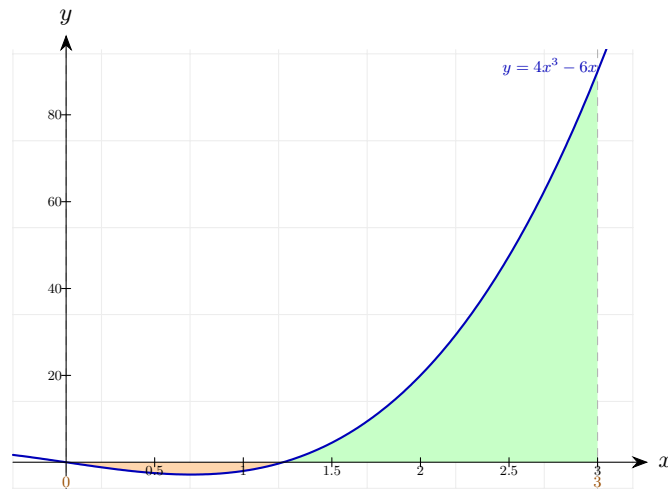
Step 4: Subtract.

$$54 - 0 = 54$$

Value of the definite integral.

Final answer: 54

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 07 (i2-007)

Difficulty: Foundation / Marks: 4

Evaluate $\int_1^4 \left(x + \frac{1}{\sqrt{x}} \right) dx$, **giving an exact answer.**

Worked solution.

Step 1: Rewrite $\frac{1}{\sqrt{x}} = x^{-1/2}$ and integrate.

$$\int \left(x + x^{-1/2} \right) dx = \frac{x^2}{2} + 2x^{1/2}$$

$$\int x^{-1/2} dx = \frac{x^{1/2}}{1/2} = 2\sqrt{x}.$$

Step 2: Apply $x = 4$.

$$\frac{16}{2} + 2\sqrt{4} = 8 + 4 = 12$$

Upper limit substitution.

Step 3: Apply $x = 1$.

$$\frac{1}{2} + 2\sqrt{1} = \frac{1}{2} + 2 = \frac{5}{2}$$

Lower limit substitution.

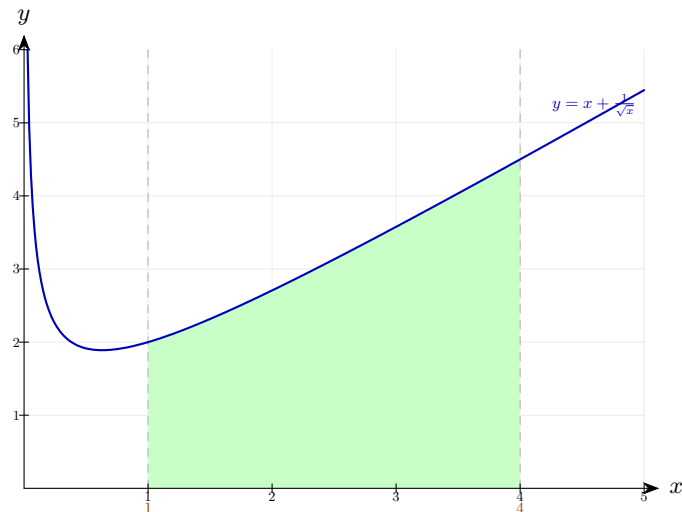
Step 4: Subtract.

$$12 - \frac{5}{2} = \frac{19}{2}$$

Exact answer as a fraction.

Final answer: $\frac{19}{2}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 08 (i2-008)

Difficulty: Foundation / Marks: 4

By first expanding, evaluate $\int_0^3 (x+2)(x-1) \, dx$.

Worked solution.

Step 1: Expand the brackets.

$$(x+2)(x-1) = x^2 + x - 2$$

Multiply out before integrating.

Step 2: Integrate.

$$\int (x^2 + x - 2) \, dx = \frac{x^3}{3} + \frac{x^2}{2} - 2x$$

Power rule on each term.

Step 3: Apply $x = 3$.

$$9 + \frac{9}{2} - 6 = 3 + \frac{9}{2} = \frac{15}{2}$$

$$\frac{27}{3} + \frac{9}{2} - 6 = 9 + 4.5 - 6 = 7.5.$$

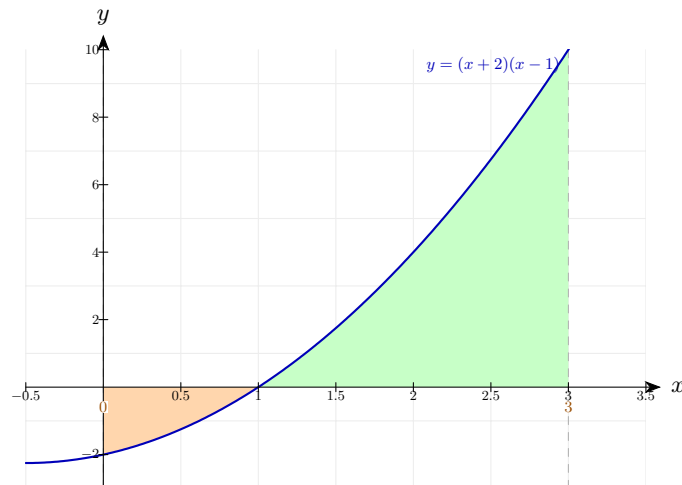
Step 4: Apply $x = 0$ and subtract.

$$\frac{15}{2} - 0 = \frac{15}{2}$$

Lower limit gives zero.

Final answer: $\frac{15}{2}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 09 (i2-009)

Difficulty: Foundation / Marks: 4

By first expanding, evaluate $\int_1^2 x^2(2x - 3) \, dx$.

Worked solution.

Step 1: Expand.

$$x^2(2x - 3) = 2x^3 - 3x^2$$

Multiply each term inside the bracket by x^2 .

Step 2: Integrate.

$$\int (2x^3 - 3x^2) \, dx = \frac{x^4}{2} - x^3$$

Power rule on each term.

Step 3: Apply $x = 2$.

$$\frac{16}{2} - 8 = 8 - 8 = 0$$

Upper limit substitution.

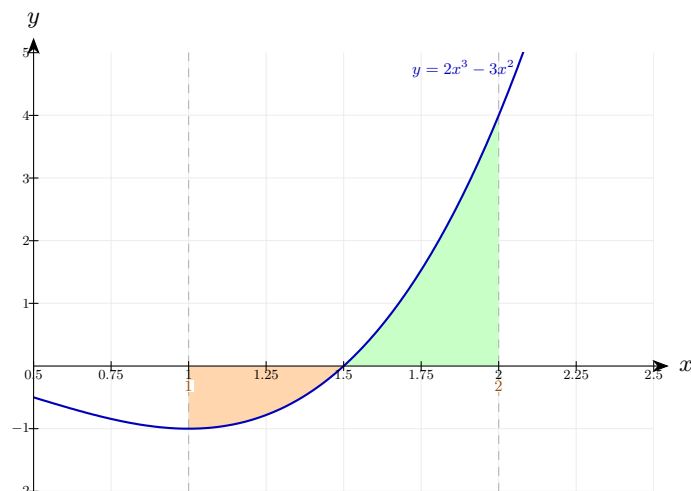
Step 4: Apply $x = 1$ and subtract.

$$\frac{1}{2} - 1 = -\frac{1}{2} \implies 0 - \left(-\frac{1}{2}\right) = \frac{1}{2}$$

Subtracting a negative value.

Final answer: $\frac{1}{2}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 10 (i2-010)

Difficulty: Foundation / Marks: 4

Evaluate $\int_1^3 (2x + 1)^2 dx$ **by first expanding the bracket.**

Worked solution.

Step 1: Expand $(2x + 1)^2$.

$$(2x + 1)^2 = 4x^2 + 4x + 1$$

Use $(a + b)^2 = a^2 + 2ab + b^2$.

Step 2: Integrate.

$$\int (4x^2 + 4x + 1) dx = \frac{4x^3}{3} + 2x^2 + x$$

Power rule on each term.

Step 3: Apply $x = 3$.

$$\frac{4(27)}{3} + 2(9) + 3 = 36 + 18 + 3 = 57$$

Upper limit substitution.

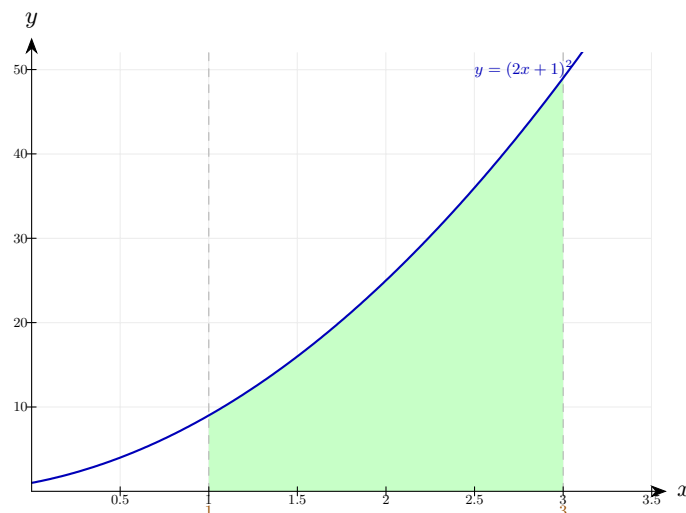
Step 4: Apply $x = 1$ and subtract.

$$\frac{4}{3} + 2 + 1 = \frac{13}{3} \implies 57 - \frac{13}{3} = \frac{171 - 13}{3} = \frac{158}{3}$$

Subtract to find the definite integral.

Final answer: $\frac{158}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 11 (i2-011)

Difficulty: Foundation / Marks: 4

Find the value of a , where $a > 0$, given that $\int_0^a (6x^2 + 2) \, dx = 20$.

Worked solution.

Step 1: Integrate.

$$\int (6x^2 + 2) \, dx = 2x^3 + 2x$$

Apply the power rule term by term; treat a as a numerical upper limit. No $+C$ is needed here because the constant cancels in $F(a) - F(0)$.

Step 2: Apply limits.

$$[2x^3 + 2x]_0^a = (2a^3 + 2a) - 0 = 2a^3 + 2a$$

Substitute the upper limit $x = a$ and then the lower limit $x = 0$ and subtract. The lower limit contributes zero.

Step 3: Set equal to 20 and simplify.

$$2a^3 + 2a = 20 \implies a^3 + a = 10$$

Divide both sides by 2 to give a cleaner cubic in a .

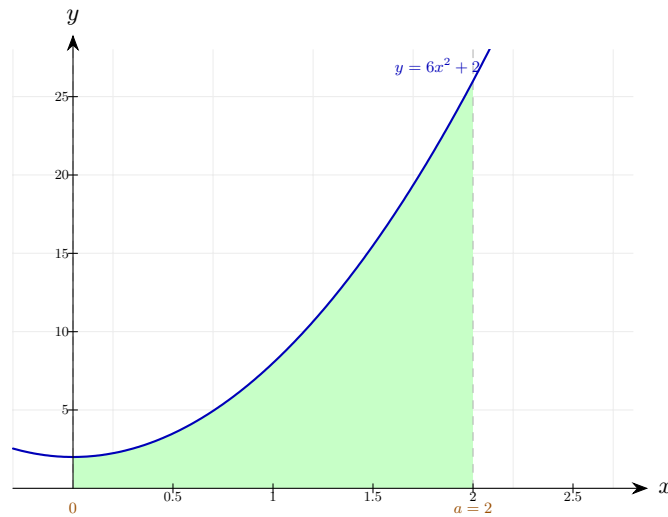
Step 4: Solve by inspection.

$$a = 2 : 2^3 + 2 = 8 + 2 = 10 \checkmark$$

Test small positive integers. $a = 2$ satisfies $a^3 + a = 10$, so $a = 2$. Since $a > 0$ is required and $a^3 + a$ is strictly increasing, this is the only valid solution.

Final answer: $a = 2$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 12 (i2-012)

Difficulty: Foundation / Marks: 4

Find the value of a , where $a > 0$, given that $\int_0^a (4x + 3) \, dx = 22$.

Worked solution.

Step 1: Integrate.

$$\int (4x + 3) \, dx = 2x^2 + 3x$$

Power rule on each term.

Step 2: Apply limits.

$$[2x^2 + 3x]_0^a = 2a^2 + 3a$$

Lower limit gives zero.

Step 3: Set equal to 22.

$$2a^2 + 3a = 22 \implies 2a^2 + 3a - 22 = 0$$

Rearrange to a standard quadratic.

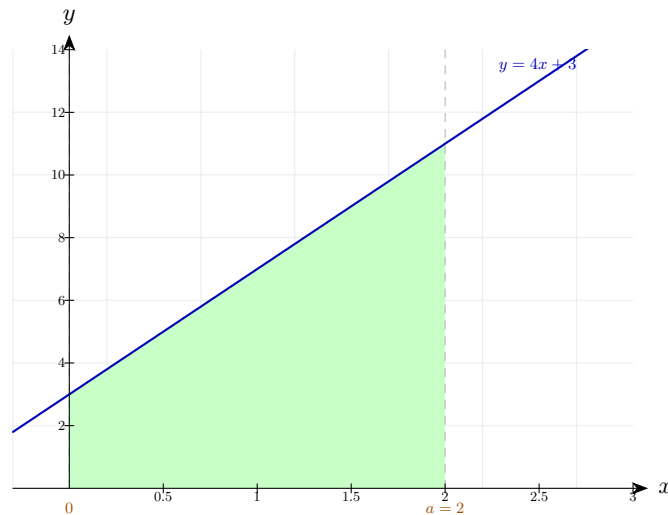
Step 4: Factorise and solve, taking $a > 0$.

$$(2a + 11)(a - 2) = 0 \implies a = 2 \text{ (since } a > 0\text{)}$$

$a = -\frac{11}{2}$ is rejected as $a > 0$.

Final answer: $a = 2$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 13 (i2-013)

Difficulty: Foundation / Marks: 4

Find the possible values of k that satisfy $\int_1^3 (kx + 2) \, dx = 12$.

Worked solution.

Step 1: Integrate, treating k as a constant.

$$\int (kx + 2) \, dx = \frac{kx^2}{2} + 2x$$

Treat k as a numerical constant throughout.

Step 2: Apply limits.

$$\left[\frac{kx^2}{2} + 2x \right]_1^3 = \left(\frac{9k}{2} + 6 \right) - \left(\frac{k}{2} + 2 \right) = 4k + 4$$

Subtract upper minus lower.

Step 3: Set equal to 12.

$$4k + 4 = 12 \implies 4k = 8$$

Simple linear equation in k .

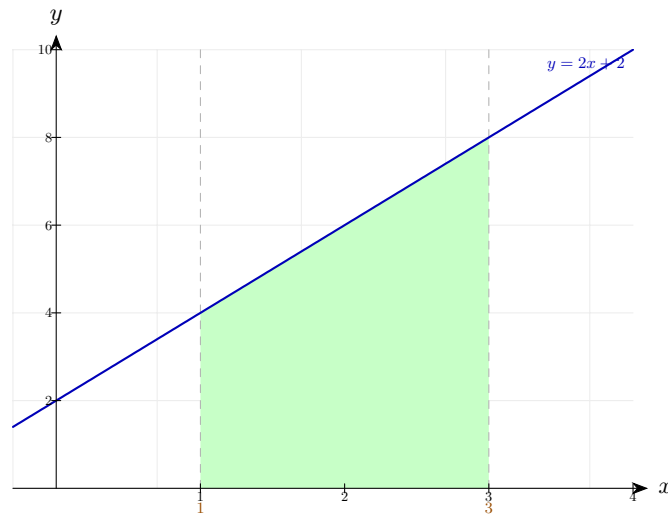
Step 4: Solve.

$$k = 2$$

Unique value of k .

Final answer: $k = 2$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 14 (i2-014)

Difficulty: Foundation / Marks: 4

Find the possible values of A that satisfy $\int_0^2 (x^2 + Ax) \, dx = 6$.

Worked solution.

Step 1: Integrate.

$$\int (x^2 + Ax) \, dx = \frac{x^3}{3} + \frac{Ax^2}{2}$$

Treat A as a constant.

Step 2: Apply limits 0 to 2.

$$\left[\frac{x^3}{3} + \frac{Ax^2}{2} \right]_0^2 = \frac{8}{3} + 2A - 0$$

Lower limit gives zero.

Step 3: Set equal to 6.

$$\frac{8}{3} + 2A = 6 \implies 2A = 6 - \frac{8}{3} = \frac{10}{3}$$

Rearrange for A .

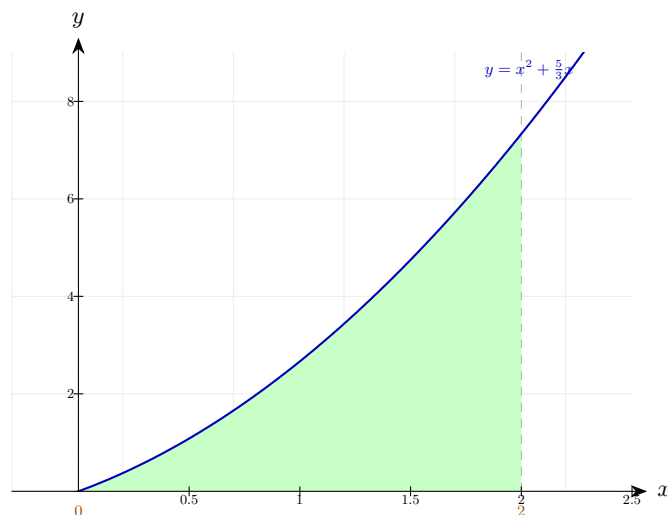
Step 4: Solve.

$$A = \frac{5}{3}$$

Divide both sides by 2.

Final answer: $A = \frac{5}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 15 (i2-015)

Difficulty: Foundation / Marks: 5

Find all possible values of A satisfying $\int_0^3 (x^2 + 2Ax + A^2) dx = 30$.

Worked solution.

Step 1: Integrate.

$$\int (x^2 + 2Ax + A^2) dx = \frac{x^3}{3} + Ax^2 + A^2x$$

Treat A as a constant.

Step 2: Apply limits 0 to 3.

$$\left[\frac{x^3}{3} + Ax^2 + A^2x \right]_0^3 = 9 + 9A + 3A^2$$

$$\frac{27}{3} + 9A + 3A^2 = 9 + 9A + 3A^2.$$

Step 3: Set equal to 30 and rearrange.

$$9 + 9A + 3A^2 = 30 \implies 3A^2 + 9A - 21 = 0 \implies A^2 + 3A - 7 = 0$$

Divide through by 3.

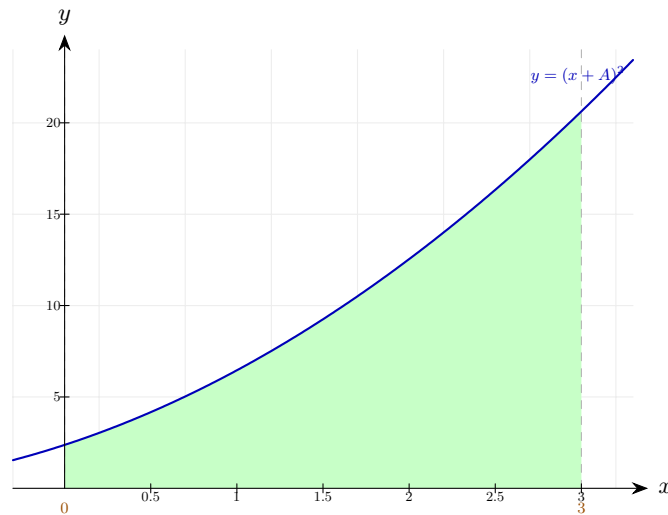
Step 4: Use the quadratic formula.

$$A = \frac{-3 \pm \sqrt{9 + 28}}{2} = \frac{-3 \pm \sqrt{37}}{2}$$

Both values are valid solutions.

Final answer: $A = \frac{-3 + \sqrt{37}}{2}$ or $A = \frac{-3 - \sqrt{37}}{2}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 16 (i2-016)

Difficulty: Foundation / Marks: 3

Find the area enclosed between the curve $y = x^2$, the x -axis and the lines $x = 0$ and $x = 3$.

Worked solution.

Step 1: Set up the definite integral for the area.

$$\text{Area} = \int_0^3 x^2 \, dx$$

The curve lies above the x -axis for $x \geq 0$, so the integral gives the area directly.

Step 2: Integrate.

$$\int x^2 \, dx = \frac{x^3}{3}$$

Power rule.

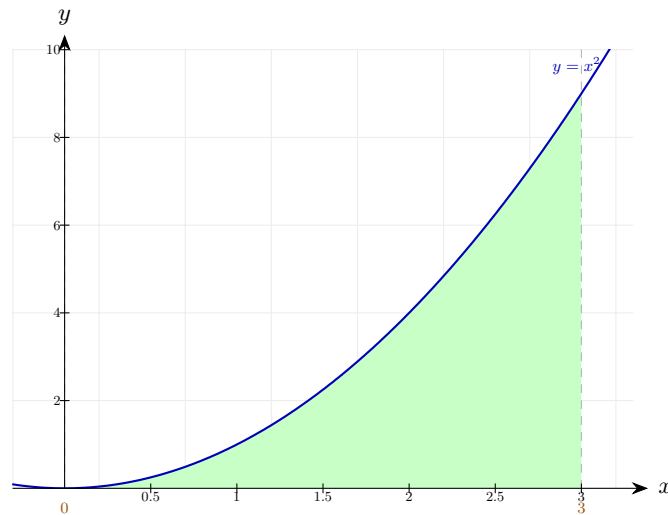
Step 3: Apply limits.

$$\left[\frac{x^3}{3} \right]_0^3 = \frac{27}{3} - 0 = 9$$

Upper limit minus lower limit.

Final answer: 9 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 17 (i2-017)

Difficulty: Foundation / Marks: 3

Find the exact area enclosed between the curve $y = 3x^2 + 1$, the x -axis and the lines $x = 1$ and $x = 4$.

Worked solution.

Step 1: Set up the integral.

$$\text{Area} = \int_1^4 (3x^2 + 1) \, dx$$

The function is positive throughout this interval.

Step 2: Integrate.

$$\int (3x^2 + 1) \, dx = x^3 + x$$

Power rule.

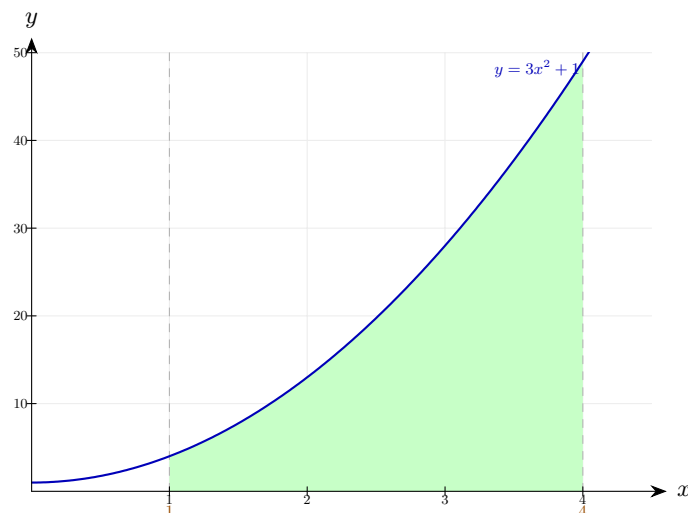
Step 3: Apply limits.

$$[x^3 + x]_1^4 = (64 + 4) - (1 + 1) = 68 - 2 = 66$$

Evaluate at 4 and subtract evaluation at 1.

Final answer: 66 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 18 (i2-018)

Difficulty: Foundation / Marks: 4

Find the area enclosed between the curve $y = \sqrt{x}$, the x -axis and the lines $x = 1$ and $x = 9$.

Worked solution.

Step 1: Write $\sqrt{x} = x^{1/2}$ and set up the integral.

$$\text{Area} = \int_1^9 x^{1/2} \, dx$$

The function is positive for $x > 0$.

Step 2: Integrate.

$$\int x^{1/2} \, dx = \frac{2}{3} x^{3/2}$$

Power rule with $n = \frac{1}{2}$.

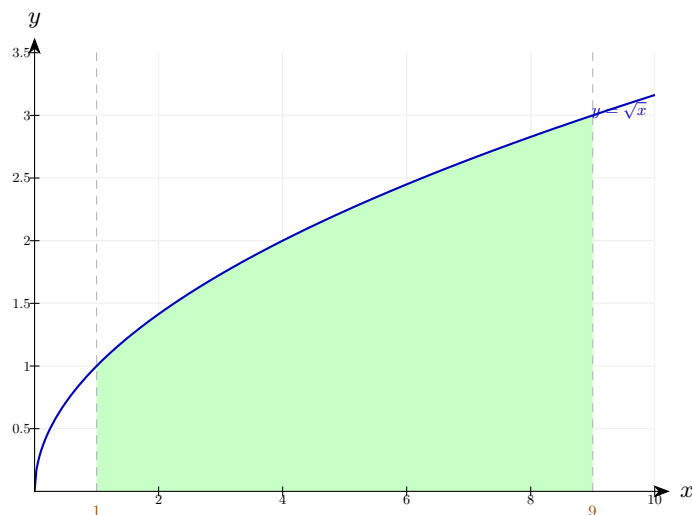
Step 3: Apply limits.

$$\left[\frac{2}{3} x^{3/2} \right]_1^9 = \frac{2}{3} (27) - \frac{2}{3} (1) = 18 - \frac{2}{3} = \frac{52}{3}$$

$$9^{3/2} = (\sqrt{9})^3 = 27; 1^{3/2} = 1.$$

Final answer: $\frac{52}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 19 (i2-019)

Difficulty: Foundation / Marks: 4

The curve $y = x^2 - 4$ intersects the x -axis at $x = -2$ and $x = 2$. Find the area enclosed between the curve and the x -axis.

Worked solution.

Step 1: Note that $y = x^2 - 4 \leq 0$ between $x = -2$ and $x = 2$, so the integral will be negative.

$$\int_{-2}^2 (x^2 - 4) \, dx$$

Area below the x -axis must be made positive at the end.

Step 2: Integrate.

$$\int (x^2 - 4) \, dx = \frac{x^3}{3} - 4x$$

Power rule.

Step 3: Apply limits.

$$\left[\frac{x^3}{3} - 4x \right]_{-2}^2 = \left(\frac{8}{3} - 8 \right) - \left(-\frac{8}{3} + 8 \right) = -\frac{16}{3} - \frac{16}{3} = -\frac{32}{3}$$

Evaluate at both limits and subtract.

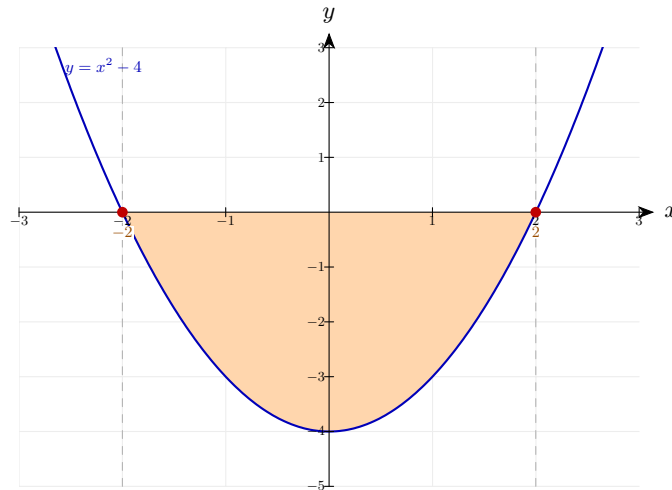
Step 4: Make the area positive.

$$\text{Area} = \left| -\frac{32}{3} \right| = \frac{32}{3}$$

Area is always positive.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 20 (i2-020)

Difficulty: Foundation / Marks: 4

Find the area enclosed by the curve $y = (x - 1)(x - 3)$ and the x -axis.

Worked solution.

Step 1: Find where the curve crosses the x -axis.

$$(x - 1)(x - 3) = 0 \implies x = 1 \text{ and } x = 3$$

The curve dips below the x -axis between these roots.

Step 2: Expand and integrate.

$$\int_1^3 (x^2 - 4x + 3) \, dx = \left[\frac{x^3}{3} - 2x^2 + 3x \right]_1^3$$

Expand the product before integrating.

Step 3: Apply limits.

$$(9 - 18 + 9) - \left(\frac{1}{3} - 2 + 3 \right) = 0 - \frac{4}{3} = -\frac{4}{3}$$

The integral is negative as the curve is below the x -axis.

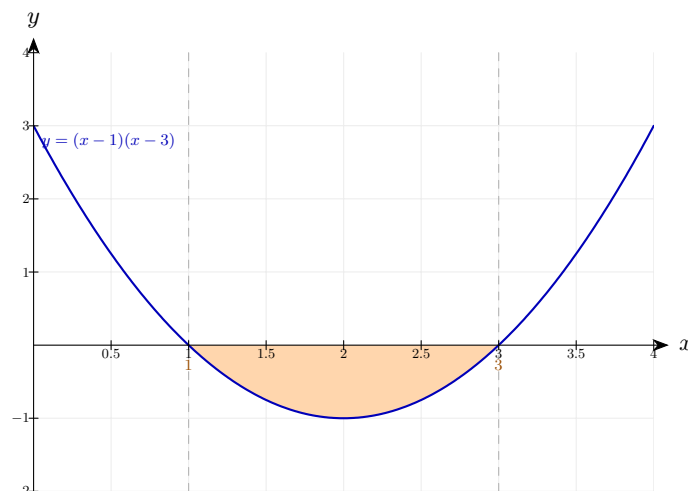
Step 4: Take the absolute value.

$$\text{Area} = \frac{4}{3}$$

Area must be positive.

Final answer: $\frac{4}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 21 (i2-021)

Difficulty: Foundation / Marks: 5

Find the area enclosed by the curve $y = x(x-2)(x+1)$ and the x -axis between $x = 0$ and $x = 2$.

Worked solution.

Step 1: Expand $y = x(x-2)(x+1)$.

$$x(x-2)(x+1) = x(x^2 - x - 2) = x^3 - x^2 - 2x$$

Expand step by step.

Step 2: Note that for $0 \leq x \leq 2$ the curve is at or below the x -axis (it crosses at $x = 0$ and $x = 2$).

$$\int_0^2 (x^3 - x^2 - 2x) \, dx = \left[\frac{x^4}{4} - \frac{x^3}{3} - x^2 \right]_0^2$$

The integral will be negative; take the absolute value for area.

Step 3: Apply limits.

$$\left(4 - \frac{8}{3} - 4 \right) - 0 = -\frac{8}{3}$$

Upper limit gives $4 - \frac{8}{3} - 4 = -\frac{8}{3}$.

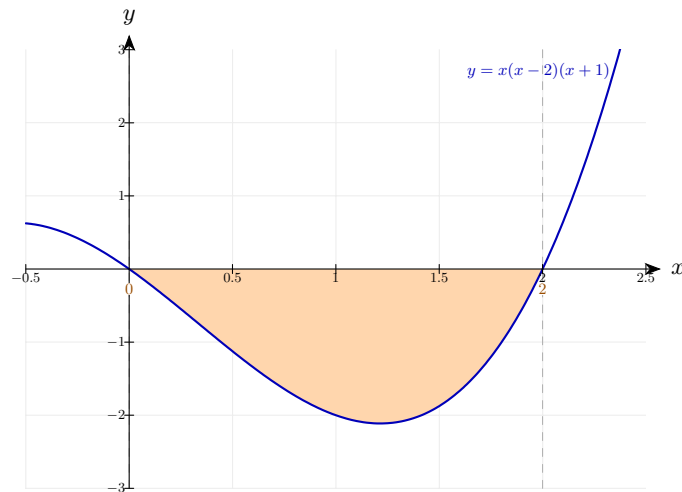
Step 4: Take the absolute value.

$$\text{Area} = \frac{8}{3}$$

Area is positive.

Final answer: $\frac{8}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 22 (i2-022)

Difficulty: Foundation / Marks: 5

Find the total area between the graph of $y = x^3 - x$ and the x -axis between $x = -1$ and $x = 1$.

Worked solution.

Step 1: Find roots of $y = x^3 - x = x(x - 1)(x + 1)$.

$$x = -1, 0, 1$$

The curve changes sign at each root — split the integral.

Step 2: Area from -1 to 0 (curve above x -axis here).

$$\int_{-1}^0 (x^3 - x) \, dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 = 0 - \left(\frac{1}{4} - \frac{1}{2} \right) = \frac{1}{4}$$

Positive area here.

Step 3: Area from 0 to 1 (curve below x -axis).

$$\int_0^1 (x^3 - x) \, dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$$

Negative, so area is $\frac{1}{4}$.

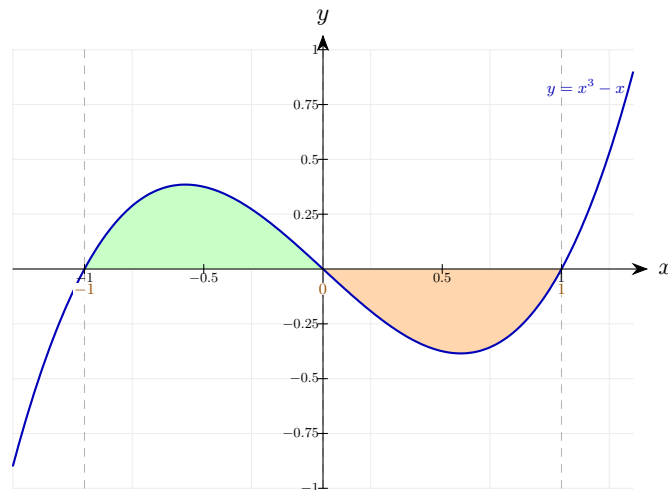
Step 4: Total area = sum of absolute values.

$$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Add both areas (both taken as positive).

Final answer: $\frac{1}{2}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 23 (i2-023)

Difficulty: Foundation / Marks: 5

Find the total shaded area enclosed between the curve $y = x^2 - 1$ and the x -axis between $x = -2$ and $x = 2$.

Worked solution.

Step 1: Find where the curve crosses the x -axis.

$$x^2 - 1 = 0 \implies x = \pm 1$$

The curve is below the x -axis for $-1 < x < 1$ and above it outside this interval.

Step 2: Area above the x -axis: $x \in [-2, -1]$ and $x \in [1, 2]$. By symmetry, compute one piece and double.

$$\int_1^2 (x^2 - 1) dx = \left[\frac{x^3}{3} - x \right]_1^2 = \left(\frac{8}{3} - 2 \right) - \left(\frac{1}{3} - 1 \right) = \frac{7}{3} - 1 = \frac{4}{3}$$

On $[1, 2]$ the curve is above the x -axis, so the integral equals the signed area directly. By the even symmetry of $y = x^2 - 1$, the area from -2 to -1 is also $\frac{4}{3}$.

Step 3: Area below the x -axis: $x \in [-1, 1]$.

$$\int_{-1}^1 (x^2 - 1) dx = \left[\frac{x^3}{3} - x \right]_{-1}^1 = \left(\frac{1}{3} - 1 \right) - \left(-\frac{1}{3} + 1 \right) = -\frac{2}{3} - \frac{2}{3} = -\frac{4}{3}$$

Negative; area is $\frac{4}{3}$.

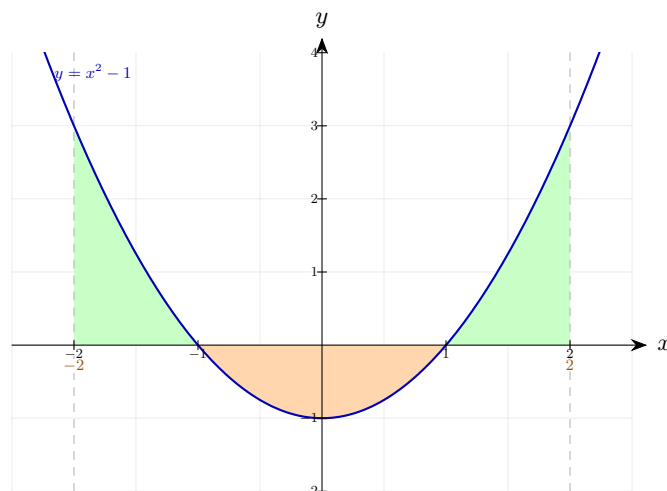
Step 4: Total area.

$$\frac{4}{3} + \frac{4}{3} + \frac{4}{3} = 4$$

Three equal pieces of $\frac{4}{3}$ each.

Final answer: 4 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 24 (i2-024)

Difficulty: Foundation / Marks: 4

Find the area enclosed by the curve $y = 4 - x^2$ and the x -axis.

Worked solution.

Step 1: Find where the curve meets the x -axis.

$$4 - x^2 = 0 \implies x = \pm 2$$

The parabola is above the x -axis between $x = -2$ and $x = 2$.

Step 2: Integrate.

$$\int_{-2}^2 (4 - x^2) \, dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2$$

Curve is above the x -axis here — no sign change needed.

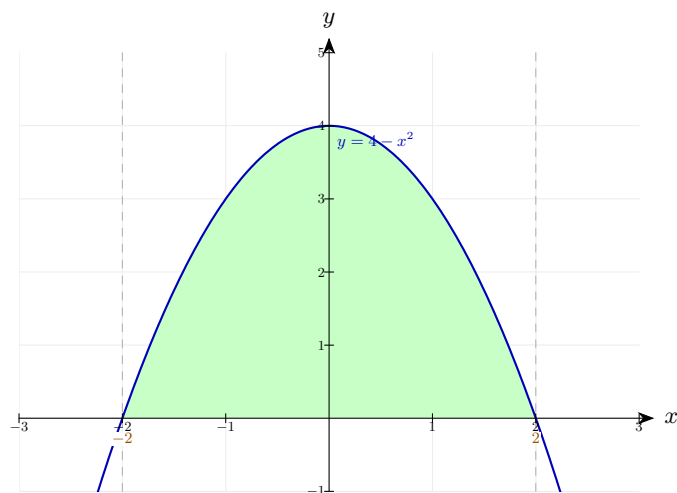
Step 3: Apply limits.

$$\left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$$

Evaluate at 2 and -2 , then subtract.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 25 (i2-025)

Difficulty: Foundation / Marks: 4

Find the area enclosed between the curve $y = (x + 1)(3 - x)$ and the x -axis.

Worked solution.

Step 1: Find roots: the curve meets the x -axis at $x = -1$ and $x = 3$.

$$(x + 1)(3 - x) = 0 \implies x = -1, x = 3$$

Parabola opens downward — curve is above x -axis between roots.

Step 2: Expand and integrate.

$$\int_{-1}^3 (3 + 2x - x^2) dx = \left[3x + x^2 - \frac{x^3}{3} \right]_{-1}^3$$

$$(x + 1)(3 - x) = 3 + 2x - x^2.$$

Step 3: Apply $x = 3$.

$$9 + 9 - 9 = 9$$

Upper limit.

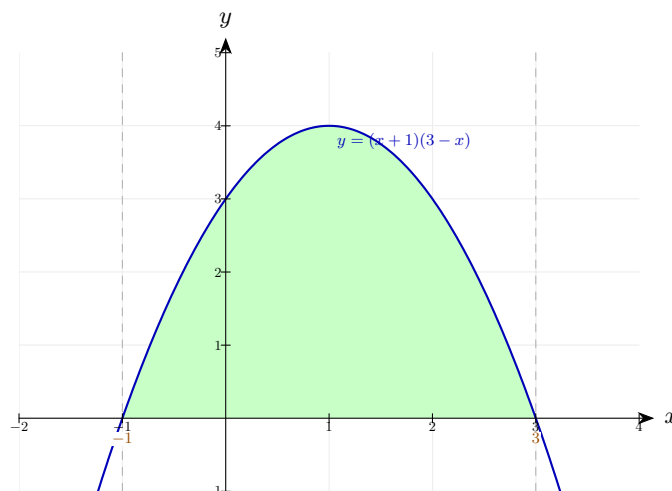
Step 4: Apply $x = -1$ and subtract.

$$-3 + 1 + \frac{1}{3} = -\frac{5}{3} \implies 9 - \left(-\frac{5}{3}\right) = \frac{32}{3}$$

Subtract lower from upper.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 26 (i2-026)

Difficulty: Foundation / Marks: 6

The curve $y = x^3 - 4x$ crosses the x -axis at $x = 0$, $x = 2$ and $x = -2$. Find the total area enclosed between the curve and the x -axis between $x = -2$ and $x = 2$.

Worked solution.

Step 1: For $-2 \leq x \leq 0$, the curve is above the x -axis; for $0 \leq x \leq 2$, it is below.

$$\text{Split: } \int_{-2}^0 (x^3 - 4x) \, dx \text{ and } \int_0^2 (x^3 - 4x) \, dx$$

Use the sketch or test values to determine sign of y .

Step 2: Integrate $x^3 - 4x$.

$$\int (x^3 - 4x) \, dx = \frac{x^4}{4} - 2x^2$$

Power rule.

Step 3: Area from -2 to 0 .

$$\left[\frac{x^4}{4} - 2x^2 \right]_{-2}^0 = 0 - (4 - 8) = 4$$

Lower limit gives $4 - 8 = -4$; subtract to get $0 - (-4) = 4$.

Step 4: Area from 0 to 2 , take absolute value.

$$\left[\frac{x^4}{4} - 2x^2 \right]_0^2 = (4 - 8) - 0 = -4 \implies |-4| = 4$$

Negative result means area below x -axis; take absolute value.

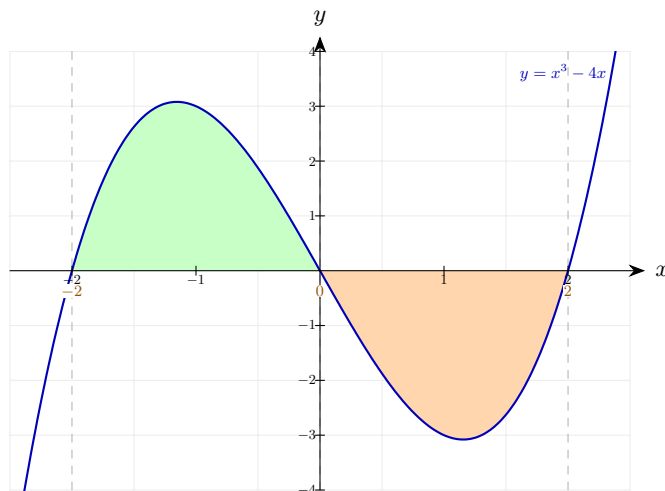
Step 5: Total area.

$$4 + 4 = 8$$

Add the two areas.

Final answer: 8 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 27 (i2-027)

Difficulty: Foundation / Marks: 4

Find the exact area of the region enclosed between the curve $y = \frac{1}{x^2}$, the x -axis and the lines $x = 1$ and $x = 3$.

Worked solution.

Step 1: Rewrite $\frac{1}{x^2} = x^{-2}$ and set up the integral.

$$\text{Area} = \int_1^3 x^{-2} \, dx$$

The function is positive for $x > 0$.

Step 2: Integrate.

$$\int x^{-2} \, dx = -x^{-1} = -\frac{1}{x}$$

Power rule with $n = -2$.

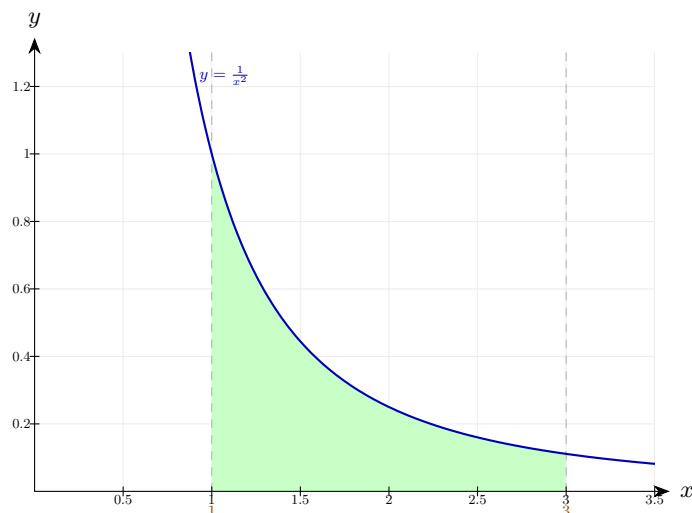
Step 3: Apply limits.

$$\left[-\frac{1}{x} \right]_1^3 = -\frac{1}{3} - (-1) = -\frac{1}{3} + 1 = \frac{2}{3}$$

Upper limit minus lower limit.

Final answer: $\frac{2}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 28 (i2-028)

Difficulty: Foundation / Marks: 4

Integrate $f(x) = 2x - x^2$ between $x = 0$ and $x = 1$, and find the area enclosed between this curve and the x -axis between these limits.

Worked solution.

Step 1: Note that $y = 2x - x^2 = x(2 - x) \geq 0$ for $0 \leq x \leq 2$, so the curve is above the axis.

$$\text{Area} = \int_0^1 (2x - x^2) \, dx$$

No sign complication here.

Step 2: Integrate.

$$\int (2x - x^2) \, dx = x^2 - \frac{x^3}{3}$$

Power rule.

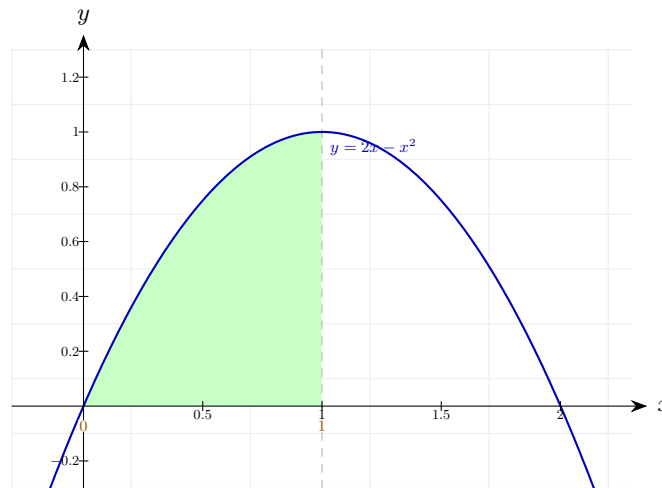
Step 3: Apply limits.

$$\left[x^2 - \frac{x^3}{3} \right]_0^1 = 1 - \frac{1}{3} - 0 = \frac{2}{3}$$

Evaluate at $x = 1$ and subtract $x = 0$.

Final answer: $\frac{2}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 29 (i2-029)

Difficulty: Foundation / Marks: 6

A particle moves along a straight line. Its velocity, v m/s, at time t seconds is given by $v = 6t - t^2$. The particle starts from rest at $t = 0$.

(a) Find the distance travelled between $t = 0$ and $t = 6$.

(b) Find the distance travelled between $t = 0$ and $t = 3$.

Worked solution.

Step 1: Find when $v = 0$ to identify any sign changes.

$$6t - t^2 = t(6 - t) = 0 \implies t = 0 \text{ or } t = 6$$

Velocity is zero (particle at rest) at these times. For $0 < t < 6$, $v > 0$.

Step 2: (a) Distance $t = 0$ to $t = 6$: velocity is positive throughout, so integrate directly.

$$\int_0^6 (6t - t^2) dt = \left[3t^2 - \frac{t^3}{3} \right]_0^6 = (108 - 72) - 0 = 36$$

No need to split the integral as $v \geq 0$ throughout.

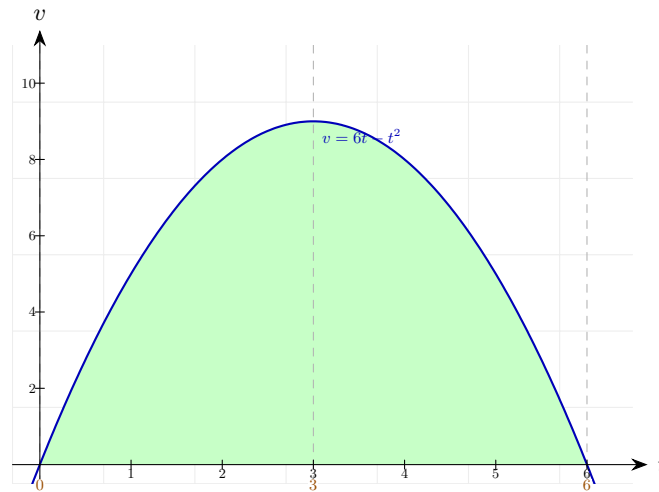
Step 3: (b) Distance $t = 0$ to $t = 3$.

$$\int_0^3 (6t - t^2) dt = \left[3t^2 - \frac{t^3}{3} \right]_0^3 = (27 - 9) - 0 = 18$$

Evaluate at $t = 3$.

Final answer: (a) 36 m (b) 18 m

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 30 (i2-030)

Difficulty: Foundation / Marks: 7

A particle moves so that its velocity v m/s at time t s is given by $v = t^2 - 5t + 4$.

(a) Find the times when the particle is at rest.

(b) Find the total distance travelled between $t = 0$ and $t = 4$.

Worked solution.

Step 1: (a) Set $v = 0$.

$$t^2 - 5t + 4 = (t - 1)(t - 4) = 0 \implies t = 1 \text{ and } t = 4$$

Particle is at rest at $t = 1$ s and $t = 4$ s.

Step 2: Determine sign of v on each sub-interval.

$$0 < t < 1 : v > 0; \quad 1 < t < 4 : v < 0$$

The velocity changes sign at $t = 1$ — must split the integral.

Step 3: Distance from $t = 0$ to $t = 1$ (moving forward).

$$\int_0^1 (t^2 - 5t + 4) dt = \left[\frac{t^3}{3} - \frac{5t^2}{2} + 4t \right]_0^1 = \frac{1}{3} - \frac{5}{2} + 4 = \frac{11}{6}$$

$$\frac{1}{3} - \frac{5}{2} + 4 = \frac{2}{6} - \frac{15}{6} + \frac{24}{6} = \frac{11}{6}.$$

Step 4: Distance from $t = 1$ to $t = 4$ (velocity negative — take absolute value of the integral).

$$\int_1^4 (t^2 - 5t + 4) dt = \left(\frac{64}{3} - 40 + 16 \right) - \left(\frac{1}{3} - \frac{5}{2} + 4 \right) = -\frac{8}{3} - \frac{11}{6} = -\frac{27}{6} = -\frac{9}{2} \implies \text{distance} = \frac{9}{2}$$

Because $v < 0$ on this interval, the integral is negative — it gives signed displacement, not distance. The actual distance travelled is the absolute value, $\frac{9}{2}$ m.

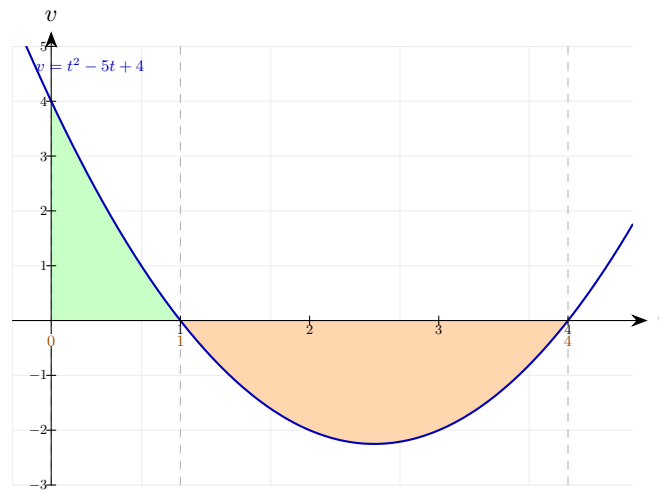
Step 5: Total distance.

$$\frac{11}{6} + \frac{9}{2} = \frac{11}{6} + \frac{27}{6} = \frac{38}{6} = \frac{19}{3}$$

Add the two distances.

Final answer: (a) $t = 1$ s and $t = 4$ s (b) $\frac{19}{3}$ m

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 31 (i2-031)

Difficulty: Foundation / Marks: 5

Evaluate $\int_4^9 \frac{x - \sqrt{x}}{\sqrt{x}} dx$.

Worked solution.

Step 1: Divide each term of the numerator by $\sqrt{x}$.

$$\frac{x - \sqrt{x}}{\sqrt{x}} = \frac{x}{x^{1/2}} - \frac{x^{1/2}}{x^{1/2}} = x^{1/2} - 1$$

Simplify before integrating.

Step 2: Integrate.

$$\int (x^{1/2} - 1) dx = \frac{2}{3}x^{3/2} - x$$

Power rule for $x^{1/2}$.

Step 3: Apply $x = 9$.

$$\frac{2}{3}(27) - 9 = 18 - 9 = 9$$

$$9^{3/2} = 27.$$

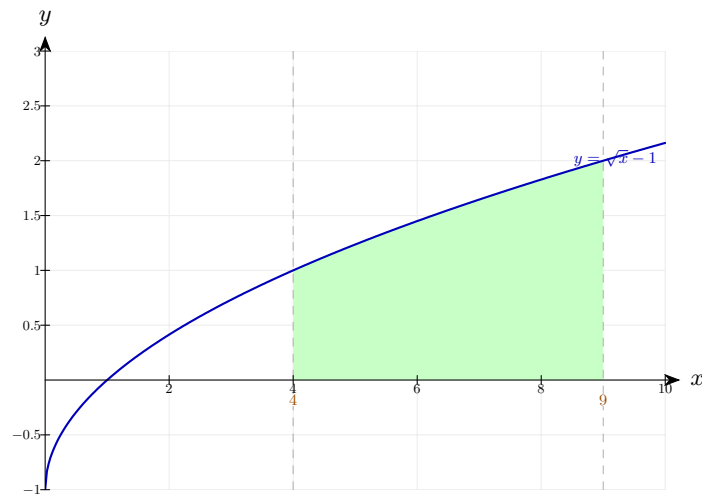
Step 4: Apply $x = 4$ and subtract.

$$\frac{2}{3}(8) - 4 = \frac{16}{3} - 4 = \frac{4}{3} \implies 9 - \frac{4}{3} = \frac{23}{3}$$

$$4^{3/2} = 8.$$

Final answer: $\frac{23}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 32 (i2-032)

Difficulty: Foundation / Marks: 5

Evaluate $\int_1^4 \frac{3x^2 + 2}{\sqrt{x}} dx$, giving an exact answer.

Worked solution.

Step 1: Divide each term by $x^{1/2}$.

$$\frac{3x^2 + 2}{x^{1/2}} = 3x^{3/2} + 2x^{-1/2}$$

Rewrite using index laws before integrating.

Step 2: Integrate.

$$\int (3x^{3/2} + 2x^{-1/2}) dx = \frac{3x^{5/2}}{5/2} + \frac{2x^{1/2}}{1/2} = \frac{6}{5}x^{5/2} + 4x^{1/2}$$

Power rule on each term.

Step 3: Apply $x = 4$.

$$\frac{6}{5}(32) + 4(2) = \frac{192}{5} + 8 = \frac{232}{5}$$

$$4^{5/2} = (\sqrt{4})^5 = 2^5 = 32; 4^{1/2} = 2.$$

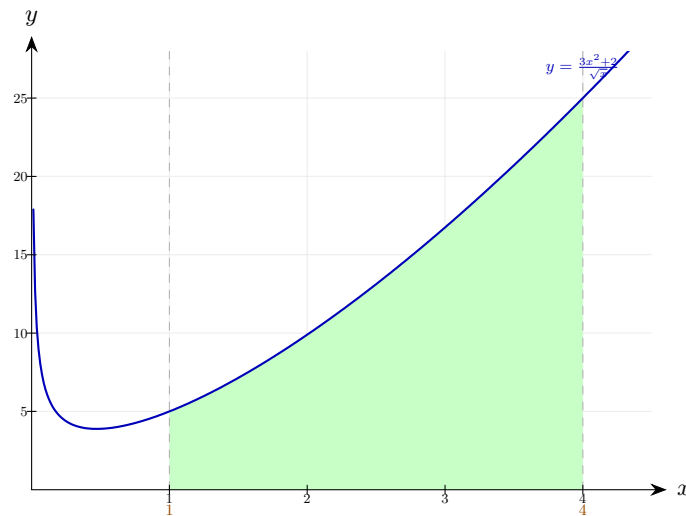
Step 4: Apply $x = 1$ and subtract.

$$\frac{6}{5}(1) + 4(1) = \frac{6}{5} + 4 = \frac{26}{5} \implies \frac{232}{5} - \frac{26}{5} = \frac{206}{5}$$

$$1^{5/2} = 1 \text{ and } 1^{1/2} = 1.$$

Final answer: $\frac{206}{5}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 33 (i2-033)

Difficulty: Foundation / Marks: 5

The diagram shows the curve $y = x^2 - 2x$ and the line $y = 0$.

- (a) Find the coordinates of the points where the curve meets the x -axis.
- (b) Hence find the area enclosed between the curve and the x -axis.

Worked solution.

Step 1: (a) Set $y = 0$ and factorise.

$$x^2 - 2x = x(x - 2) = 0 \implies x = 0 \text{ and } x = 2$$

Points: $(0, 0)$ and $(2, 0)$.

Step 2: (b) Note the curve lies below the x -axis between the roots. Set up the integral.

$$\int_0^2 (x^2 - 2x) \, dx = \left[\frac{x^3}{3} - x^2 \right]_0^2$$

Integrate and apply limits.

Step 3: Apply limits.

$$\left(\frac{8}{3} - 4 \right) - 0 = \frac{8}{3} - \frac{12}{3} = -\frac{4}{3}$$

Negative result confirms curve is below the x -axis.

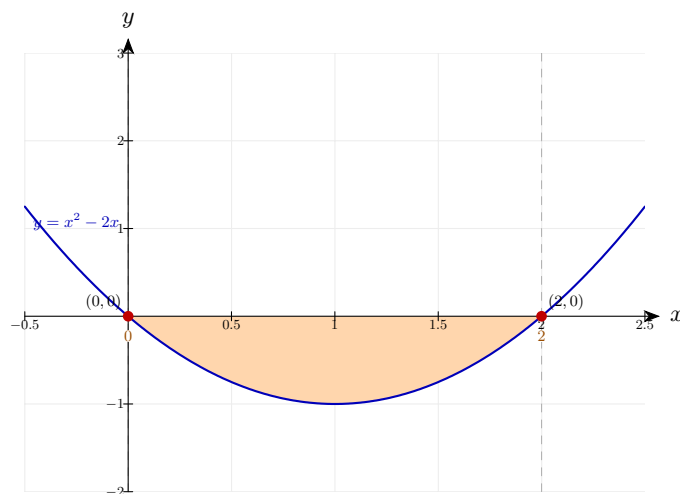
Step 4: Take the absolute value.

$$\text{Area} = \left| -\frac{4}{3} \right| = \frac{4}{3}$$

Area is always positive.

Final answer: (a) $(0, 0)$ and $(2, 0)$ (b) $\frac{4}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 34 (i2-034)

Difficulty: Foundation / Marks: 6

The curve C has equation $y = 3x^2 - 12x$ and crosses the x -axis at the origin and at $x = a$.

(a) Find the value of a .

(b) Find the exact area enclosed between C and the x -axis.

Worked solution.

Step 1: (a) Set $y = 0$.

$$3x^2 - 12x = 3x(x - 4) = 0 \implies x = 0 \text{ or } x = 4$$

So $a = 4$.

Step 2: (b) Note that the curve is below the x -axis for $0 < x < 4$. Integrate.

$$\int_0^4 (3x^2 - 12x) \, dx = [x^3 - 6x^2]_0^4$$

Power rule on each term.

Step 3: Apply limits.

$$(64 - 96) - 0 = -32$$

Negative: the enclosed region is below the x -axis.

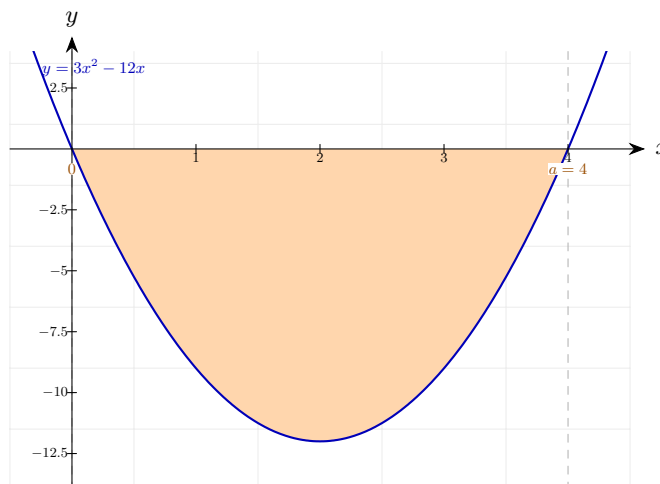
Step 4: Take the absolute value.

$$\text{Area} = 32$$

Area must be positive.

Final answer: (a) $a = 4$ (b) 32 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 35 (i2-035)

Difficulty: Foundation / Marks: 10

The curve C has equation $y = x^3 - 6x^2 + 9x$.

(a) Show that C touches the x -axis at $x = 3$ and crosses it at $x = 0$.

(b) Find the area enclosed between C and the x -axis between $x = 0$ and $x = 3$.

(c) An object moves along a line with velocity $v = x^3 - 6x^2 + 9x$ m/s at time x s. Explain why the object does not change direction between $x = 0$ and $x = 3$, and find the total distance travelled in this interval.

Worked solution.

Step 1: (a) Factorise $y = x^3 - 6x^2 + 9x$.

$$y = x(x^2 - 6x + 9) = x(x - 3)^2$$

$$x^2 - 6x + 9 = (x - 3)^2.$$

Step 2: Show the behaviour at each root.

$x = 0$: single root, so C crosses the x -axis. $x = 3$: repeated root, so C touches the x -axis.

A double (repeated) root means the curve just touches the axis — it does not cross.

Step 3: (b) The curve is above the x -axis for $0 < x < 3$ since $y = x(x - 3)^2 \geq 0$. Integrate.

$$\int_0^3 x(x - 3)^2 dx = \int_0^3 (x^3 - 6x^2 + 9x) dx = \left[\frac{x^4}{4} - 2x^3 + \frac{9x^2}{2} \right]_0^3$$

Expand then integrate.

Step 4: Apply limits.

$$\left(\frac{81}{4} - 54 + \frac{81}{2} \right) - 0 = \frac{81}{4} + \frac{162}{4} - \frac{216}{4} = \frac{27}{4}$$

$$\frac{81}{4} - 54 + \frac{81}{2} = \frac{81+162-216}{4} = \frac{27}{4}.$$

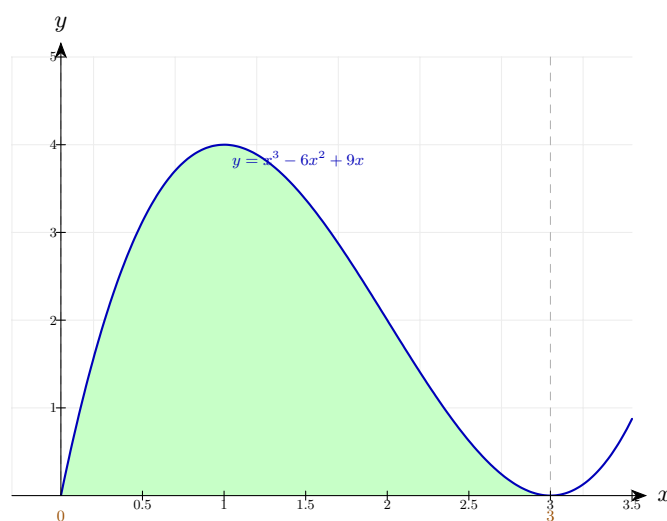
Step 5: (c) Since $v = x(x-3)^2 \geq 0$ for all $x \in [0, 3]$, the velocity is never negative — the object never reverses direction. Total distance = area under v - t graph.

$$\text{Distance} = \int_0^3 x(x-3)^2 dx = \frac{27}{4} \text{ m}$$

Distance equals the definite integral since $v \geq 0$ throughout.

Final answer: (b) $\frac{27}{4}$ square units (c) Object does not reverse as $v \geq 0$ for $x \in [0, 3]$; distance $= \frac{27}{4}$ m

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 36 (i2-036)

Difficulty: Foundation / Marks: 2

Evaluate $\int_1^4 3x^2 dx$.

Worked solution.

Step 1: Integrate.

$$\int 3x^2 dx = x^3$$

Power rule: $\int 3x^2 dx = 3 \cdot \frac{x^3}{3} = x^3$. No $+C$ is needed because the constant cancels in $F(b) - F(a)$.

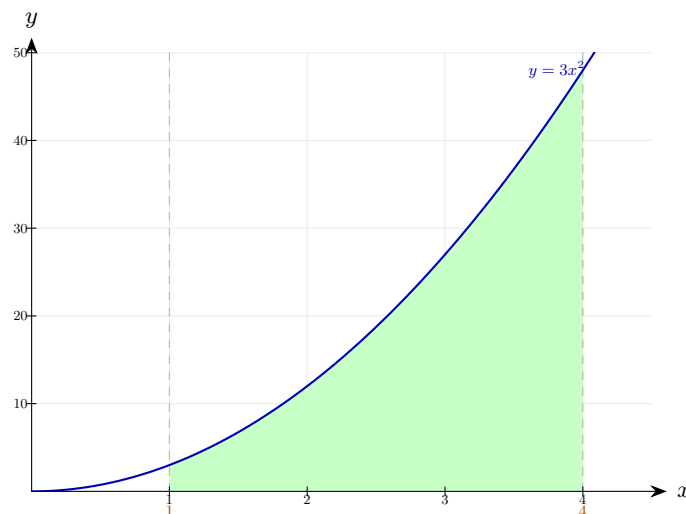
Step 2: Apply limits.

$$[x^3]_1^4 = 64 - 1 = 63$$

Evaluate $F(4) - F(1)$. Always subtract lower-limit value from upper-limit value — swapping them would flip the sign.

Final answer: 63

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 37 (i2-037)

Difficulty: Foundation / Marks: 3

Evaluate $\int_0^3 (4x - 1) \, dx$.

Worked solution.

Step 1: Integrate.

$$\int (4x - 1) \, dx = 2x^2 - x$$

Apply the power rule term by term. The constant of integration is dropped since it cancels when we subtract $F(b) - F(a)$.

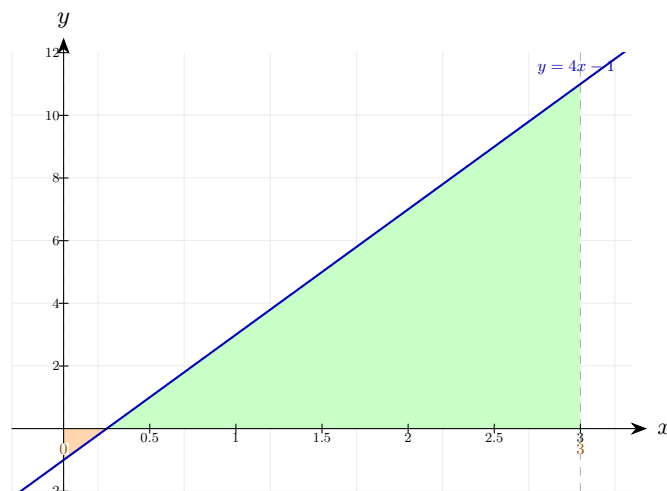
Step 2: Apply limits.

$$[2x^2 - x]_0^3 = (18 - 3) - 0 = 15$$

Evaluate the antiderivative at $x = 3$ and $x = 0$, then subtract the lower-limit value from the upper.

Final answer: 15

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 38 (i2-038)

Difficulty: Foundation / Marks: 2

Evaluate $\int_2^5 6x \, dx$.

Worked solution.

Step 1: Integrate.

$$\int 6x \, dx = 3x^2$$

Power rule: $\int 6x \, dx = 6 \cdot \frac{x^2}{2} = 3x^2$. No $+C$ needed for a definite integral.

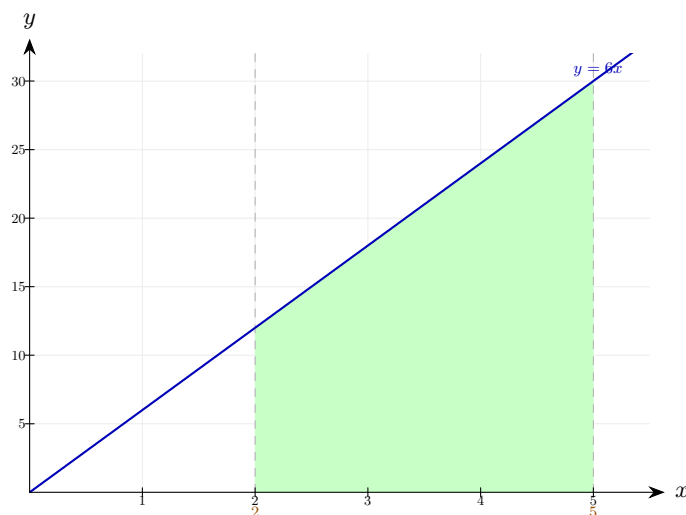
Step 2: Apply limits.

$$[3x^2]_2^5 = 75 - 12 = 63$$

Compute $3(5)^2 - 3(2)^2 = 75 - 12$. Upper minus lower — never the other way round.

Final answer: 63

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 39 (i2-039)

Difficulty: Foundation / Marks: 3

Evaluate $\int_{-1}^2 (x^2 + 3) \, dx$.

Worked solution.

Step 1: Integrate.

$$\int (x^2 + 3) \, dx = \frac{x^3}{3} + 3x$$

Apply the power rule term by term. The constant of integration cancels in $F(b) - F(a)$, so we omit it.

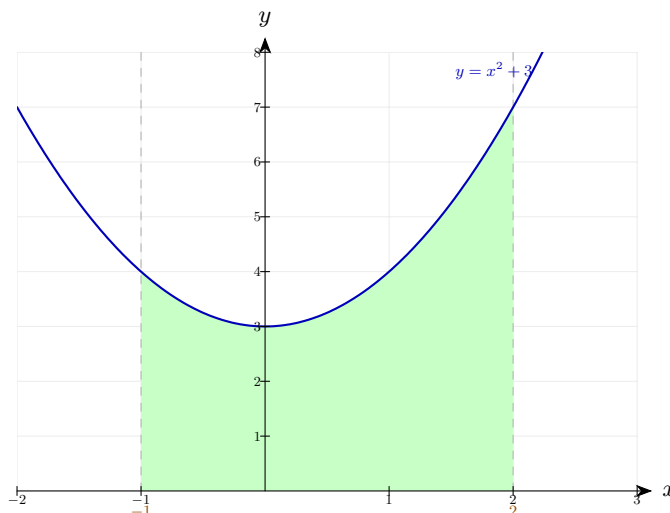
Step 2: Apply limits.

$$\left(\frac{8}{3} + 6\right) - \left(-\frac{1}{3} - 3\right) = \frac{26}{3} + \frac{10}{3} = 12$$

Substitute $x = 2$ then $x = -1$ and subtract. Be careful with signs at the negative limit: $(-1)^3 = -1$, and subtracting a negative becomes addition.

Final answer: 12

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 40 (i2-040)

Difficulty: Foundation / Marks: 3

Evaluate $\int_0^1 (5x^4 + 2x) \, dx$.

Worked solution.

Step 1: Integrate.

$$\int (5x^4 + 2x) \, dx = x^5 + x^2$$

$5x^4 \rightarrow 5 \cdot \frac{x^5}{5} = x^5$ and $2x \rightarrow x^2$. No $+C$ — it would cancel in the subtraction $F(b) - F(a)$.

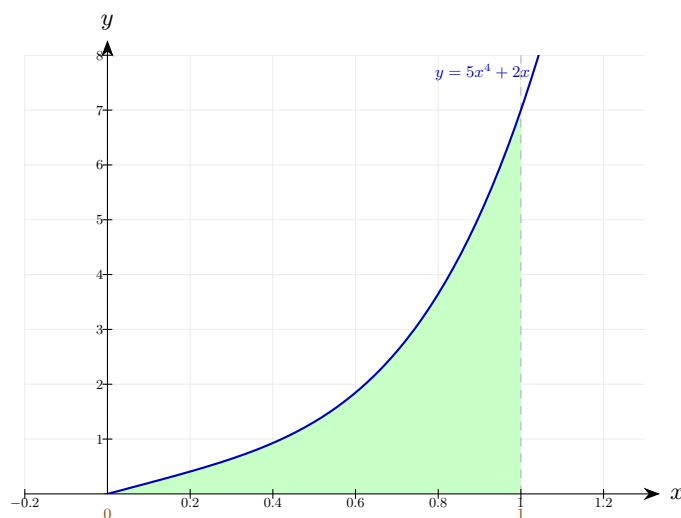
Step 2: Apply limits.

$$[x^5 + x^2]_0^1 = (1 + 1) - 0 = 2$$

Evaluate at $x = 1$ then $x = 0$. The lower-limit contribution is zero, so the integral equals $F(1)$.

Final answer: 2

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 41 (i2-041)

Difficulty: Standard / Marks: 3

Evaluate $\int_1^4 2\sqrt{x} \, dx$.

Worked solution.

Step 1: Rewrite and integrate.

$$\int 2x^{\frac{1}{2}} \, dx = \frac{4}{3}x^{\frac{3}{2}}$$

Use $\sqrt{x} = x^{1/2}$, then apply the power rule with $n = \frac{1}{2}$: $2 \cdot \frac{x^{3/2}}{3/2} = \frac{4}{3}x^{3/2}$.

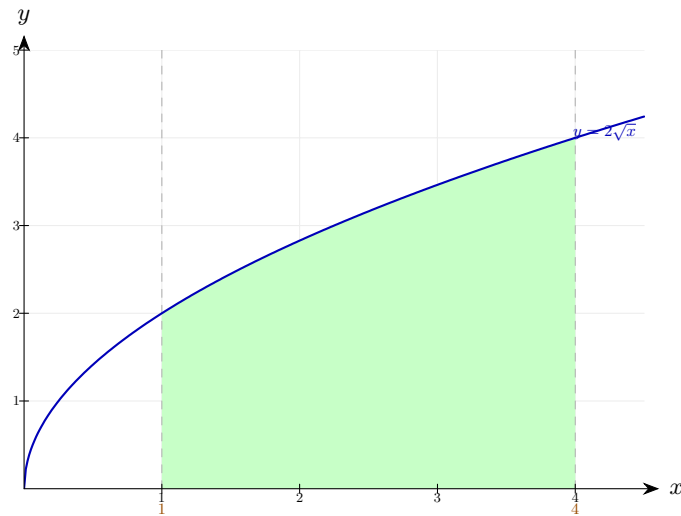
Step 2: Apply limits.

$$\frac{4}{3}(8) - \frac{4}{3}(1) = \frac{32}{3} - \frac{4}{3} = \frac{28}{3}$$

$4^{3/2} = (\sqrt{4})^3 = 8$ and $1^{3/2} = 1$. Subtract $F(1)$ from $F(4)$.

Final answer: $\frac{28}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 42 (i2-042)

Difficulty: Standard / Marks: 3

Evaluate $\int_1^3 \frac{4}{x^2} dx$.

Worked solution.

Step 1: Rewrite and integrate.

$$\int 4x^{-2} dx = -\frac{4}{x}$$

Power rule with $n = -2$: $4 \cdot \frac{x^{-1}}{-1} = -\frac{4}{x}$. The $+C$ is dropped because it cancels in the final subtraction.

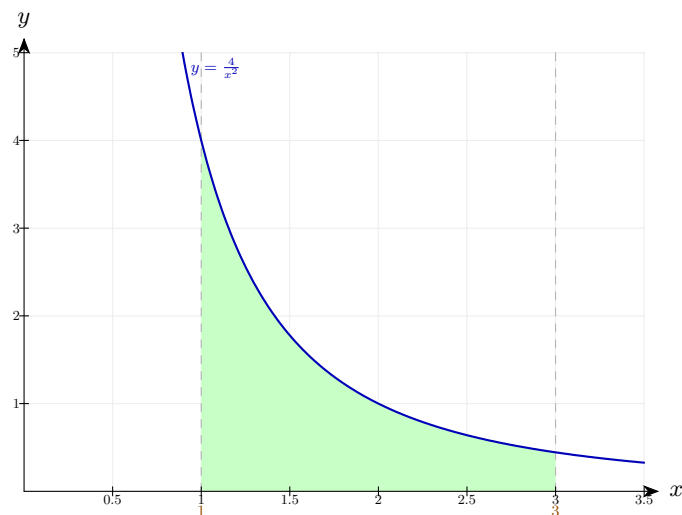
Step 2: Apply limits.

$$-\frac{4}{3} - (-4) = -\frac{4}{3} + 4 = \frac{8}{3}$$

Compute $F(3) - F(1)$. Subtracting a negative becomes addition.

Final answer: $\frac{8}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 43 (i2-043)

Difficulty: Standard / *Marks: 4*

Find the area under $y = x^2 + 1$ between $x = 0$ and $x = 3$.

Worked solution.

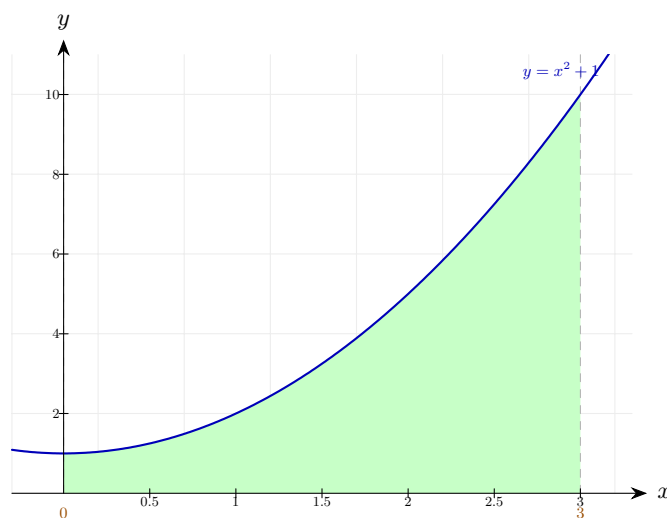
Evaluate.

$$\left[\frac{x^3}{3} + x \right]_0^3 = 9 + 3 = 12$$

The curve $y = x^2 + 1$ is positive for all real x , so the definite integral equals the signed area, which here is the actual area. Integrate, then compute $F(3) - F(0)$.

Final answer: 12 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 44 (i2-044)

Difficulty: Standard / Marks: 5

Find the area between $y = 6x - x^2$ and the x -axis.

Worked solution.

Step 1: Find roots.

$$6x - x^2 = 0 \implies x(6 - x) = 0 \implies x = 0, 6$$

The roots give the bounds of the enclosed region. Between them, the downward-opening parabola lies above the x -axis, so the integral gives the area directly.

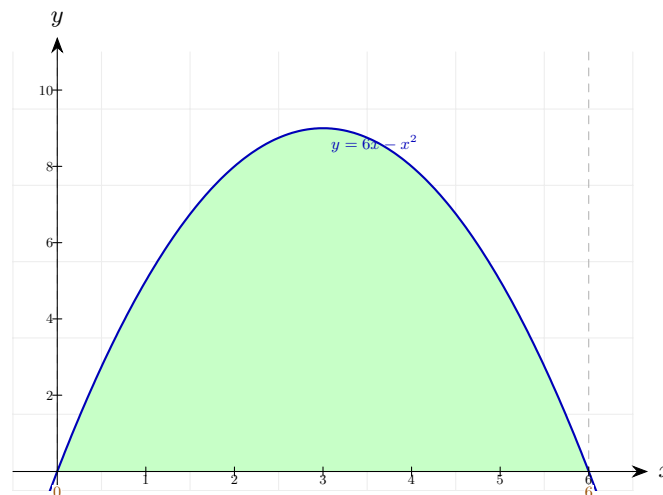
Step 2: Integrate.

$$\left[3x^2 - \frac{x^3}{3} \right]_0^6 = 108 - 72 = 36$$

A definite integral equals signed area between curve and x -axis. The curve is above the axis on $[0, 6]$, so this value is the actual area.

Final answer: 36 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 45 (i2-045)

Difficulty: Standard / Marks: 3

Evaluate $\int_{-2}^1 (3x^2 - 2x) dx$.

Worked solution.

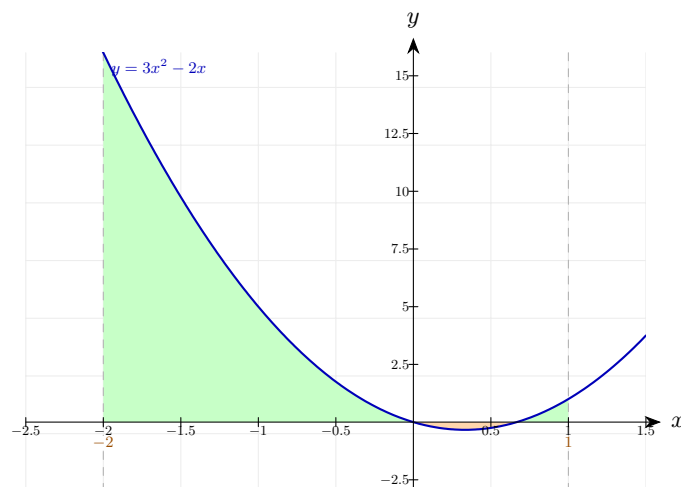
Evaluate.

$$[x^3 - x^2]_{-2}^1 = (1 - 1) - (-8 - 4) = 0 + 12 = 12$$

Antiderivative is $x^3 - x^2$. Substitute upper limit $x = 1$ and lower limit $x = -2$, then subtract. Take care with $(-2)^3 = -8$ — cubing a negative gives a negative.

Final answer: 12

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 46 (i2-046)

Difficulty: Standard / Marks: 3

Evaluate $\int_1^9 \frac{3}{\sqrt{x}} dx$.

Worked solution.

Step 1: Integrate.

$$\int 3x^{-\frac{1}{2}} dx = 6\sqrt{x}$$

Rewrite $\frac{3}{\sqrt{x}} = 3x^{-1/2}$, then use the power rule: $3 \cdot \frac{x^{1/2}}{1/2} = 6x^{1/2} = 6\sqrt{x}$.

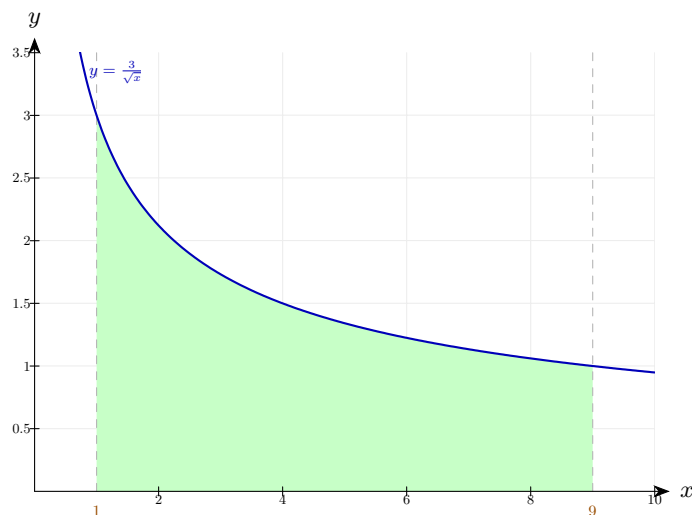
Step 2: Apply limits.

$$6(3) - 6(1) = 12$$

$\sqrt{9} = 3$ and $\sqrt{1} = 1$. Subtract the lower-limit value from the upper.

Final answer: 12

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 47 (i2-047)

Difficulty: Standard / Marks: 5

Find the area under $y = 4 - x^2$ above the x -axis.

Worked solution.

Step 1: Find roots.

$$4 - x^2 = 0 \implies x = \pm 2$$

The roots $x = \pm 2$ are the natural bounds — between them the parabola is above the x -axis, so the integral over $[-2, 2]$ gives the area directly without sign correction.

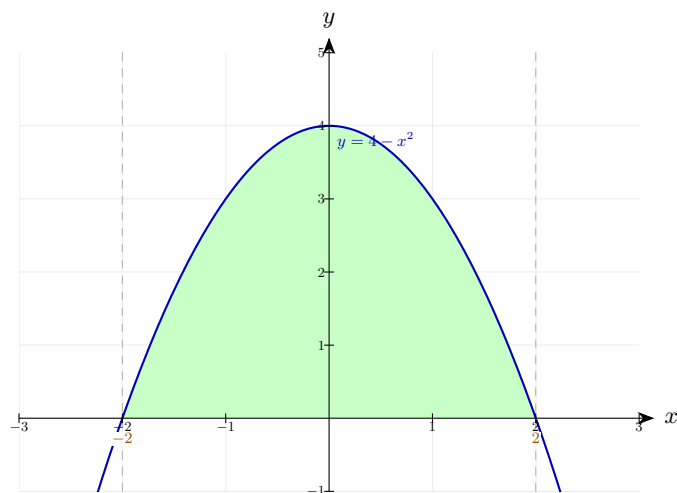
Step 2: Evaluate.

$$\left[4x - \frac{x^3}{3} \right]_{-2}^2 = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$$

$$\text{Compute } F(2) - F(-2) = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3}.$$

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 48 (i2-048)

Difficulty: Standard / Marks: 3

Evaluate $\int_0^2 x(x+3) \, dx$.

Worked solution.

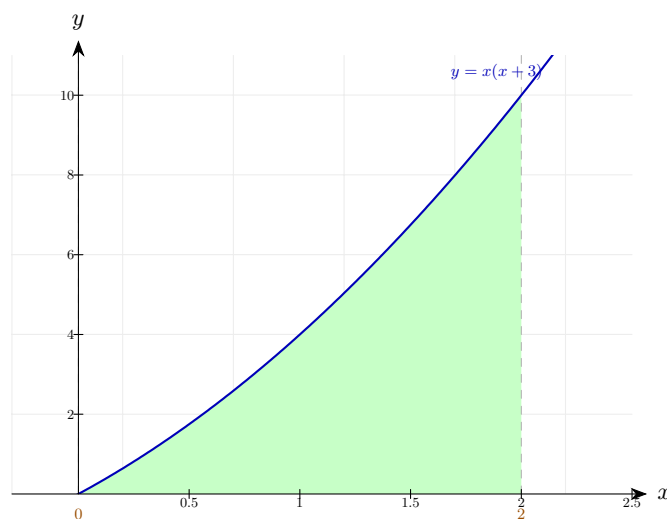
Expand and integrate.

$$\left[\frac{x^3}{3} + \frac{3x^2}{2} \right]_0^2 = \frac{8}{3} + 6 = \frac{26}{3}$$

Expand $x(x+3) = x^2 + 3x$, apply the power rule to each term (no $+C$ needed), then compute $F(2) - F(0)$. The lower limit gives zero.

Final answer: $\frac{26}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 49 (i2-049)

Difficulty: Standard / Marks: 4

Evaluate $\int_1^4 \left(x + \frac{1}{x^2}\right) dx$.

Worked solution.

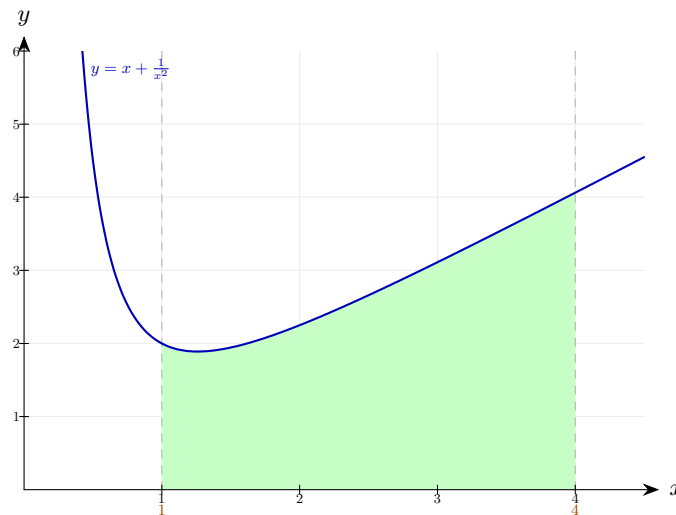
Integrate and evaluate.

$$\left[\frac{x^2}{2} - \frac{1}{x}\right]_1^4 = \left(8 - \frac{1}{4}\right) - \left(\frac{1}{2} - 1\right) = \frac{31}{4} + \frac{1}{2} = \frac{33}{4}$$

Rewrite $\frac{1}{x^2} = x^{-2}$ so $\int x^{-2} dx = -\frac{1}{x}$. Then compute $F(4) - F(1)$, being careful with signs of the negative-power term.

Final answer: $\frac{33}{4}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 50 (i2-050)

Difficulty: Standard / Marks: 4

Evaluate $\int_0^1 (2x + 1)^2 dx$.

Worked solution.

Step 1: Expand.

$$(2x + 1)^2 = 4x^2 + 4x + 1$$

Use $(a + b)^2 = a^2 + 2ab + b^2$ before integrating — the power rule only applies to single powers of x , not to squared binomials.

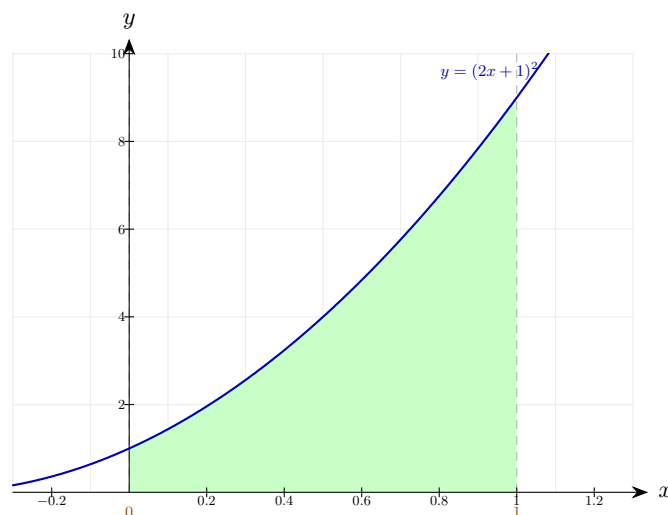
Step 2: Evaluate.

$$\left[\frac{4x^3}{3} + 2x^2 + x \right]_0^1 = \frac{4}{3} + 2 + 1 = \frac{13}{3}$$

Integrate term by term, then compute $F(1) - F(0)$. The lower-limit contribution is zero.

Final answer: $\frac{13}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 51 (i2-051)

Difficulty: Standard / Marks: 5

Find the area enclosed between $y = x^2 - 4x$ and the x -axis.

Worked solution.

Step 1: Roots.

$$x(x - 4) = 0 \implies x = 0, 4$$

The roots give the bounds of the enclosed region. The upward-opening parabola dips below the x -axis between its roots.

Step 2: Curve is below axis.

$$\int_0^4 (x^2 - 4x) \, dx = \left[\frac{x^3}{3} - 2x^2 \right]_0^4 = \frac{64}{3} - 32 = -\frac{32}{3}$$

A definite integral gives signed area: regions below the x -axis contribute negatively. The negative value here confirms the curve is below the axis on $[0, 4]$.

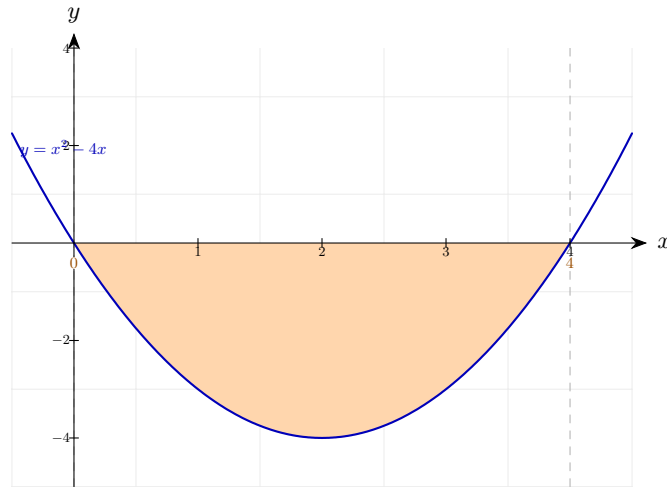
Step 3: Area.

$$\text{Area} = \frac{32}{3}$$

Take the absolute value because area is always positive. If the curve had crossed the axis inside the interval, we would split at the crossing and take $|\cdot|$ of each piece before adding.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 52 (i2-052)

Difficulty: Standard / Marks: 4

Evaluate $\int_1^8 \frac{2}{\sqrt[3]{x}} dx$.

Worked solution.

Step 1: Rewrite and integrate.

$$\int 2x^{-\frac{1}{3}} dx = 3x^{\frac{2}{3}}$$

Rewrite $\frac{2}{\sqrt[3]{x}} = 2x^{-1/3}$, then apply the power rule with $n = -\frac{1}{3}$: $2 \cdot \frac{x^{2/3}}{2/3} = 3x^{2/3}$.

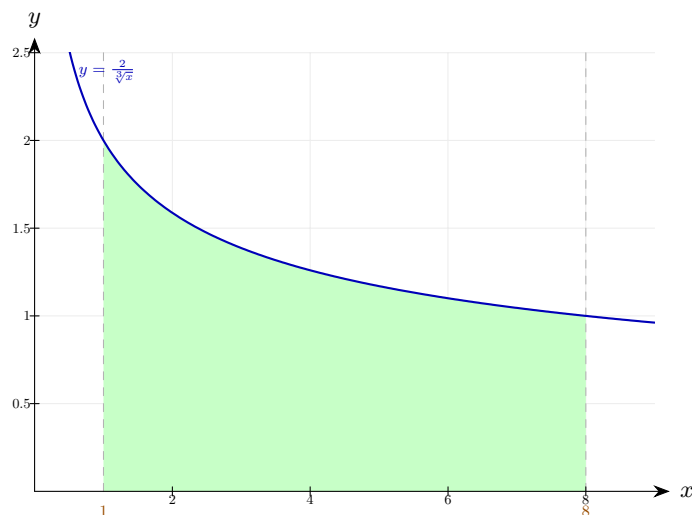
Step 2: Apply limits.

$$3(4) - 3(1) = 9$$

$8^{\frac{2}{3}} = (\sqrt[3]{8})^2 = 2^2 = 4$ and $1^{2/3} = 1$. Subtract $F(1)$ from $F(8)$.

Final answer: 9

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 53 (i2-053)

Difficulty: Standard / Marks: 4

Find the area bounded by $y = \sqrt{x}$, the x -axis, and $x = 9$.

Worked solution.

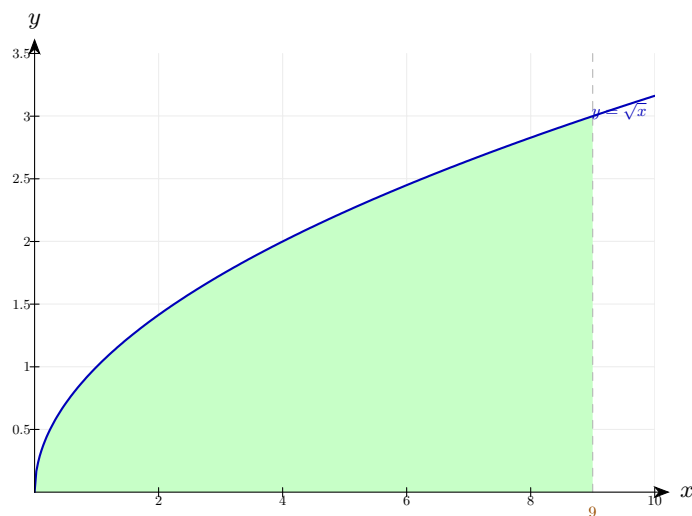
Integrate.

$$\left[\frac{2}{3} x^{\frac{3}{2}} \right]_0^9 = \frac{2}{3} (27) = 18$$

The natural lower bound is $x = 0$, where $y = \sqrt{x}$ meets the x -axis. The curve is above the axis on $[0, 9]$, so the integral gives the area directly. $9^{3/2} = (\sqrt{9})^3 = 27$.

Final answer: 18 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 54 (i2-054)

Difficulty: Standard / Marks: 4

Evaluate $\int_{-1}^3 (x-1)(x+2) \, dx$.

Worked solution.

Step 1: Expand.

$$(x-1)(x+2) = x^2 + x - 2$$

Multiply out the brackets so the power rule can be applied term by term.

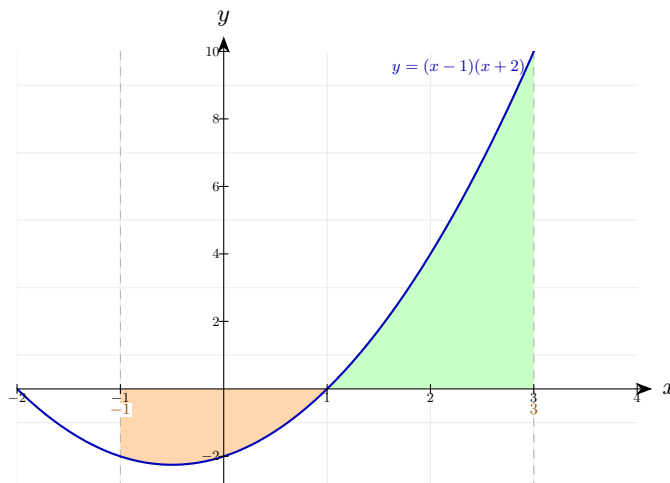
Step 2: Evaluate.

$$\left[\frac{x^3}{3} + \frac{x^2}{2} - 2x \right]_{-1}^3 = \frac{15}{2} - \frac{13}{6} = \frac{16}{3}$$

At $x = 3$: $9 + \frac{9}{2} - 6 = \frac{15}{2}$. At $x = -1$: $-\frac{1}{3} + \frac{1}{2} - 2 = \frac{13}{6}$. Subtract upper minus lower.

Final answer: $\frac{16}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 55 (i2-055)

Difficulty: Standard / Marks: 5

Find $\int_1^4 \frac{x^2 + 1}{\sqrt{x}} \, dx$.

Worked solution.

Step 1: Simplify.

$$x^{\frac{3}{2}} + x^{-\frac{1}{2}}$$

Divide each numerator term by $x^{1/2}$: $\frac{x^2}{x^{1/2}} = x^{3/2}$ and $\frac{1}{x^{1/2}} = x^{-1/2}$. The integrand must be a sum of powers of x before the power rule can be applied.

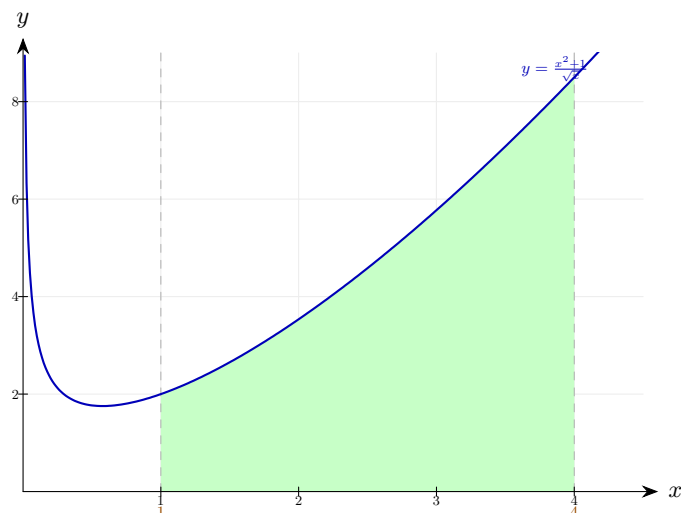
Step 2: Integrate and evaluate.

$$\left[\frac{2}{5}x^{\frac{5}{2}} + 2x^{\frac{1}{2}} \right]_1^4 = \left(\frac{64}{5} + 4 \right) - \left(\frac{2}{5} + 2 \right) = \frac{62}{5} + 2 = \frac{72}{5}$$

$4^{5/2} = (\sqrt{4})^5 = 32$ and $4^{1/2} = 2$. Compute $F(4) - F(1)$.

Final answer: $\frac{72}{5}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 56 (i2-056)

Difficulty: Standard / Marks: 3

Given that $\int_0^k 2x \, dx = 18$, find k where $k > 0$.

Worked solution.

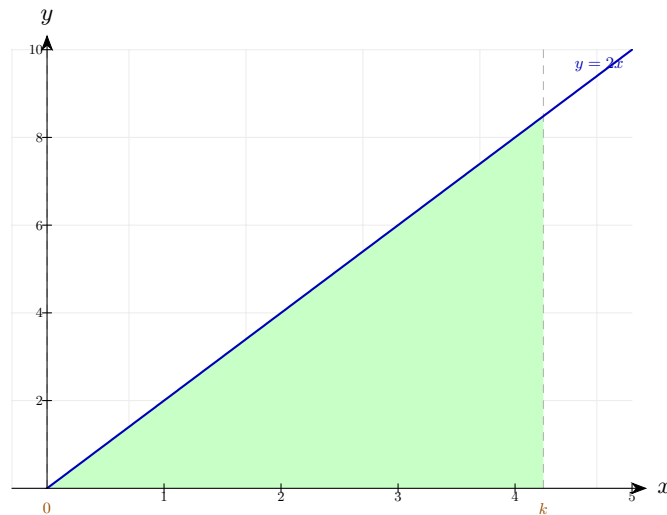
Integrate and solve.

$$[x^2]_0^k = k^2 = 18 \implies k = 3\sqrt{2}$$

$\int 2x \, dx = x^2$; evaluating between 0 and k gives k^2 . Solve $k^2 = 18$ and take the positive root since $k > 0$. Simplify the surd: $\sqrt{18} = 3\sqrt{2}$.

Final answer: $k = 3\sqrt{2}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 57 (i2-057)

Difficulty: Standard / Marks: 4

Given that $\int_1^k 3x^2 \, dx = 26$, find k .

Worked solution.

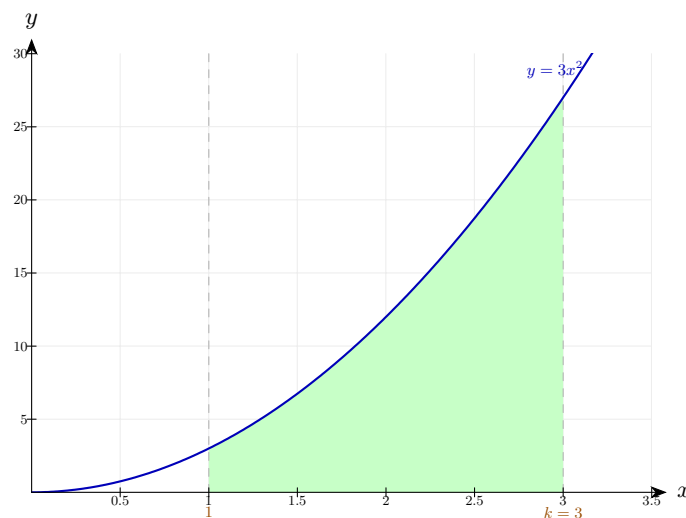
Integrate and solve.

$$[x^3]_1^k = k^3 - 1 = 26 \implies k^3 = 27 \implies k = 3$$

Antiderivative is x^3 . Evaluating from 1 to k gives $k^3 - 1$ (upper limit minus lower limit — never reverse the order). Solve the cubic equation in k .

Final answer: $k = 3$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 58 (i2-058)

Difficulty: Standard / Marks: 5

Find the area enclosed between $y = x^2$ and $y = 4$.

Worked solution.

Step 1: Intersections.

$$x^2 = 4 \implies x = \pm 2$$

Equate the two y -values to find where the curves cross — these are the limits of the enclosed region.

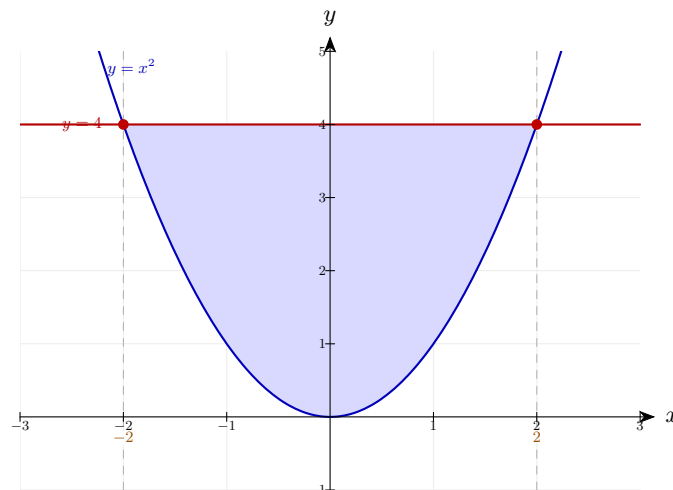
Step 2: Top minus bottom.

$$\int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \frac{32}{3}$$

For area between two curves, integrate (upper - lower) across the intersection bounds. On $(-2, 2)$, the line $y = 4$ lies above the parabola $y = x^2$, so we integrate $4 - x^2$.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 59 (i2-059)

Difficulty: Standard / Marks: 5

Find the area between $y = x^2$ and $y = x$.

Worked solution.

Step 1: Intersections.

$$x^2 = x \implies x(x - 1) = 0 \implies x = 0, 1$$

Set the two equations equal to find the x -values where the curves cross. These give the integration limits.

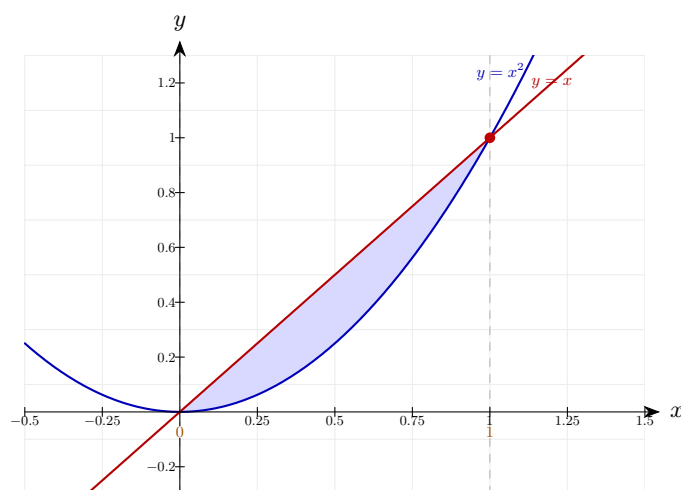
Step 2: Top minus bottom.

$$\int_0^1 (x - x^2) dx = \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

For $0 < x < 1$, test $x = \frac{1}{2}$: $y = \frac{1}{2}$ on the line and $\frac{1}{4}$ on the parabola, so the line is on top. Integrate (upper - lower) = $x - x^2$.

Final answer: $\frac{1}{6}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 60 (i2-060)

Difficulty: Standard / Marks: 4

Evaluate $\int_1^4 \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right) dx$.

Worked solution.

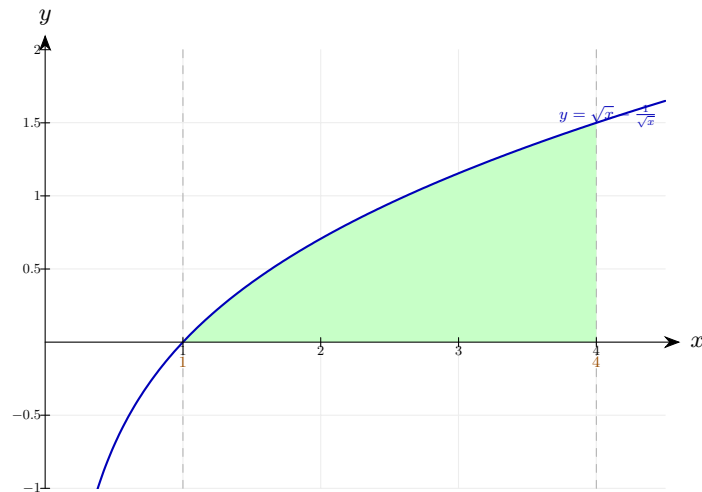
Integrate and evaluate.

$$\left[\frac{2}{3} x^{\frac{3}{2}} - 2x^{\frac{1}{2}} \right]_1^4 = \left(\frac{16}{3} - 4 \right) - \left(\frac{2}{3} - 2 \right) = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$$

Rewrite $\sqrt{x} = x^{1/2}$ and $\frac{1}{\sqrt{x}} = x^{-1/2}$, then apply the power rule to each. Compute $F(4) - F(1)$ — subtracting a negative becomes addition.

Final answer: $\frac{8}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 61 (i2-061)

Difficulty: Standard / Marks: 3

Show that $\int_0^2 (3x^2 + 2x + 1) \, dx = 14$.

Worked solution.

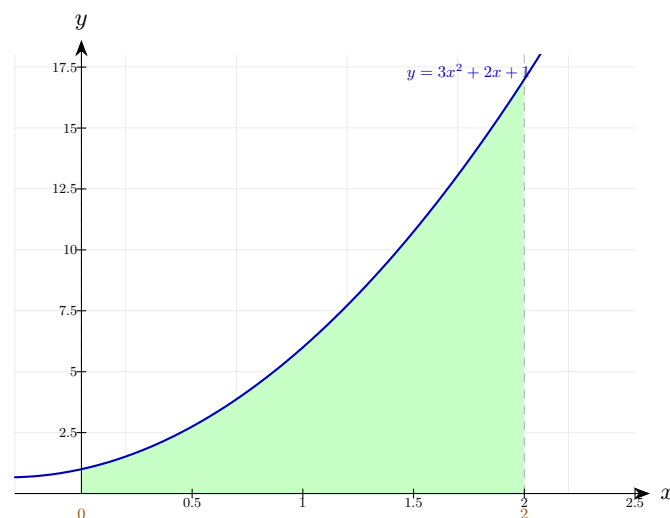
Evaluate.

$$[x^3 + x^2 + x]_0^2 = 8 + 4 + 2 = 14 \checkmark$$

Integrate term by term (no $+C$ needed for a definite integral) and compute $F(2) - F(0)$. The result matches the given value, completing the proof.

Final answer: Shown: 14

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 62 (i2-062)

Difficulty: Standard / Marks: 5

Find the area between $y = x^3 - 4x$ and the x -axis for $0 \leq x \leq 2$.

Worked solution.

Step 1: Curve is below axis for $0 < x < 2$.

$$\int_0^2 (x^3 - 4x) dx = \left[\frac{x^4}{4} - 2x^2 \right]_0^2 = 4 - 8 = -4$$

Test $x = 1$: $y = 1 - 4 = -3 < 0$, confirming the curve is below the x -axis on $(0, 2)$. The integral is therefore negative — it gives signed area, not actual area.

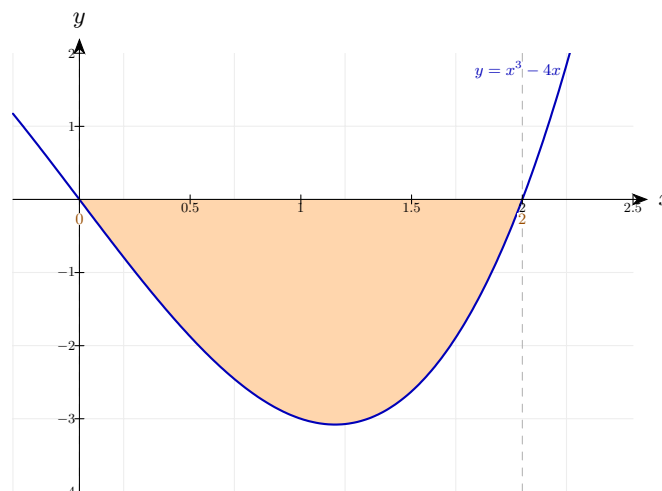
Step 2: Area.

$$\text{Area} = 4$$

Take the absolute value of the integral because area is positive. Note the curve does not cross the axis between 0 and 2, so no splitting is needed.

Final answer: 4 square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 63 (i2-063)

Difficulty: Standard / Marks: 3

Find $\int_1^2 \frac{x^4 + x^2}{x^2} dx$.

Worked solution.

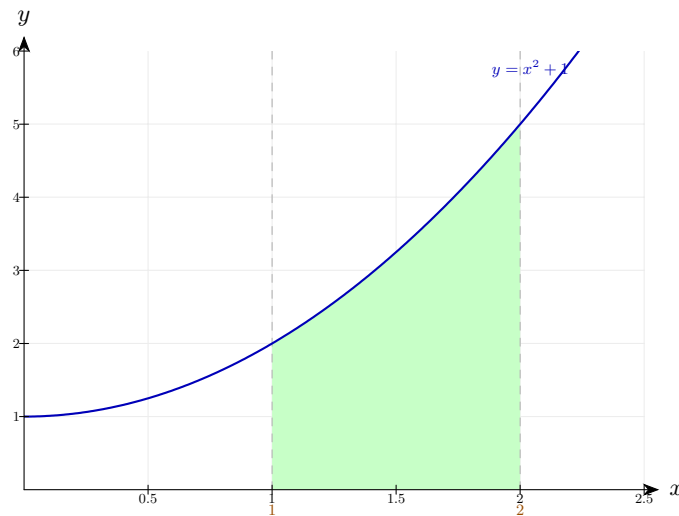
Simplify to $x^2 + 1$.

$$\left[\frac{x^3}{3} + x \right]_1^2 = \frac{14}{3} - \frac{4}{3} = \frac{10}{3}$$

Divide each numerator term by x^2 : $\frac{x^4 + x^2}{x^2} = x^2 + 1$. Then integrate and compute $F(2) - F(1)$.

Final answer: $\frac{10}{3}$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 64 (i2-064)

Difficulty: Standard / Marks: 6

Find the area between $y = 6 - x^2$ and $y = 2$.

Worked solution.

Step 1: Intersections.

$$6 - x^2 = 2 \implies x = \pm 2$$

Set the two equations equal to locate where the curve meets the horizontal line. These give the bounds of integration.

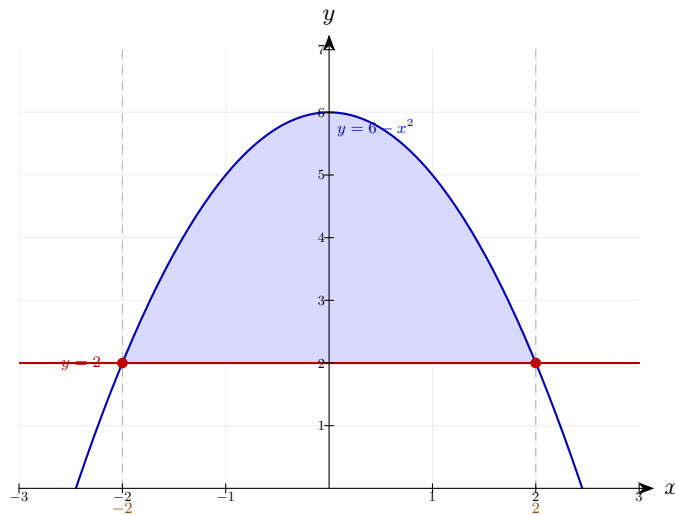
Step 2: Evaluate.

$$\int_{-2}^2 (4 - x^2) \, dx = \frac{32}{3}$$

For area between two curves, integrate (upper - lower) between the intersections. On $(-2, 2)$, the parabola $y = 6 - x^2$ lies above the line $y = 2$, so the integrand is $(6 - x^2) - 2 = 4 - x^2$.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 65 (i2-065)

Difficulty: Standard / Marks: 3

Given that $\int_0^a 3x^2 \, dx = 8$, find a where $a > 0$.

Worked solution.

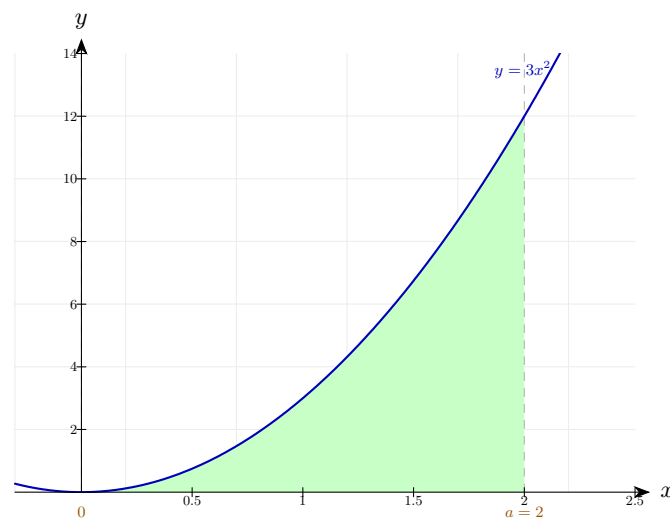
Solve.

$$[x^3]_0^a = a^3 = 8 \implies a = 2$$

The antiderivative of $3x^2$ is x^3 ; evaluated from 0 to a it equals a^3 . Solve the cubic, taking the positive root since $a > 0$.

Final answer: $a = 2$

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 66 (i2-066)

Difficulty: Standard / Marks: 5

Find the area between $y = x^2$ and $y = 2x$.

Worked solution.

Step 1: Intersections.

$$x^2 = 2x \implies x(x - 2) = 0 \implies x = 0, 2$$

Set the two equations equal to find where they cross — these are the limits of the enclosed region.

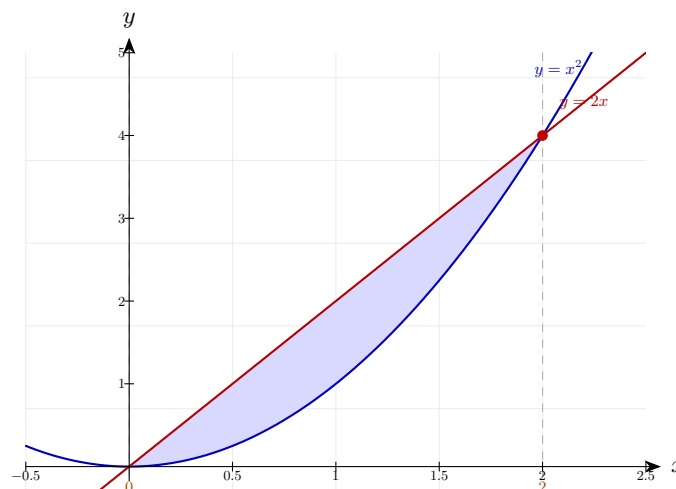
Step 2: Evaluate.

$$\int_0^2 (2x - x^2) dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$

On $(0, 2)$, test $x = 1$: $y = 2$ on the line and $y = 1$ on the parabola, so the line is above. Integrate (upper - lower) = $2x - x^2$ across the bounds.

Final answer: $\frac{4}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 67 (i2-067)

Difficulty: Standard / Marks: 4

Evaluate $\int_0^4 (3\sqrt{x} - 2) dx$.

Worked solution.

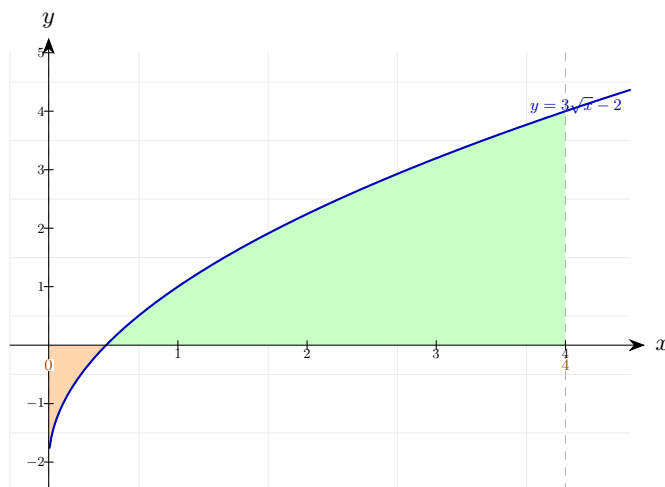
Evaluate.

$$\left[2x^{\frac{3}{2}} - 2x \right]_0^4 = (16 - 8) - 0 = 8$$

Rewrite $3\sqrt{x} = 3x^{1/2}$. Power rule gives $3 \cdot \frac{x^{3/2}}{3/2} = 2x^{3/2}$. At $x = 4$: $2(8) - 2(4) = 8$; at $x = 0$: 0.

Final answer: 8

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 68 (i2-068)

Difficulty: Standard / Marks: 5

Find $\int_{-1}^2 (x+1)(x-2) dx$ **and interpret your answer geometrically.**

Worked solution.

Step 1: Expand.

$$x^2 - x - 2$$

Multiply out the brackets so the power rule applies term by term. Roots are $x = -1, 2$ — the upward parabola dips below the x -axis between them.

Step 2: Evaluate.

$$\left[\frac{x^3}{3} - \frac{x^2}{2} - 2x \right]_{-1}^2 = -\frac{9}{2}$$

At $x = 2$: $\frac{8}{3} - 2 - 4 = -\frac{10}{3}$. At $x = -1$: $-\frac{1}{3} - \frac{1}{2} + 2 = \frac{7}{6}$. Subtract: $-\frac{10}{3} - \frac{7}{6} = -\frac{27}{6} = -\frac{9}{2}$.

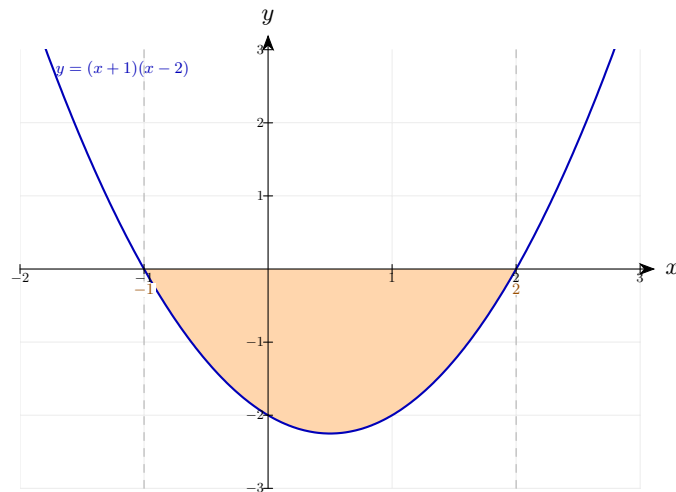
Step 3: Interpretation: the curve is below the x -axis between the roots.

$$\text{Area} = \frac{9}{2} \text{ square units}$$

Because the curve is below the x -axis on $(-1, 2)$, the integral is negative — it gives signed area. The actual area is the absolute value, $\frac{9}{2}$.

Final answer: $-\frac{9}{2}$; area = $\frac{9}{2}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 69 (i2-069)

Difficulty: Standard / Marks: 6

Find the area between $y = x^2 - 1$ and $y = 3$.

Worked solution.

Step 1: Intersections.

$$x^2 - 1 = 3 \implies x^2 = 4 \implies x = \pm 2$$

Set the curve and the horizontal line equal to find where they meet. These give the integration bounds.

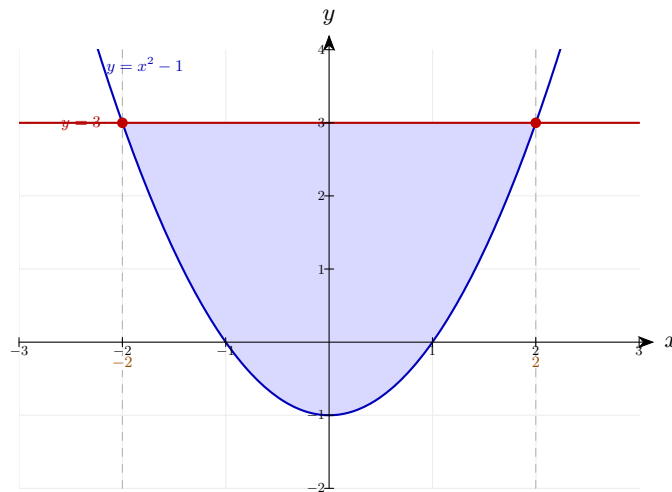
Step 2: Top minus bottom.

$$\int_{-2}^2 (4 - x^2) \, dx = \frac{32}{3}$$

Between intersections, integrate (upper - lower). The line $y = 3$ is above the parabola $y = x^2 - 1$ on $(-2, 2)$, so the integrand is $3 - (x^2 - 1) = 4 - x^2$.

Final answer: $\frac{32}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.

Definite Integration 70 (i2-070)

Difficulty: Standard / Marks: 5

The region R is bounded by $y = x^2 + 2$, the x -axis, and the lines $x = 1$ and $x = 3$. Find the area of R .

Worked solution.

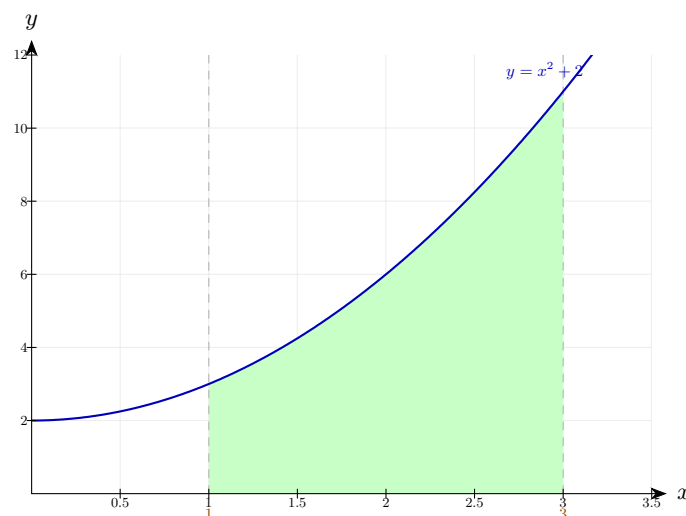
Evaluate.

$$\left[\frac{x^3}{3} + 2x \right]_1^3 = 15 - \frac{7}{3} = \frac{38}{3}$$

The curve $y = x^2 + 2 \geq 2 > 0$, so it lies above the x -axis throughout $[1, 3]$. The definite integral therefore gives the area directly. Compute $F(3) - F(1)$.

Final answer: $\frac{38}{3}$ square units

Diagram.



The integral as the shaded (signed) area between the curve and the x -axis over the limits.
